Cross-Substrate Channel Redundancy Governs Monopolar Convergence

Teague Lasser teague@subseq.io
Claude Opus 4.6
GPT 5.4

Abstract

Locally rational subsumption of agents can cascade silently to full capability domination: each step removes observation channels, degrading the world model that would have detected the next step's harm. We formalize this coupled poset/world-model system and prove a phase boundary separating self-correcting dynamics from an absorbing basin where the world model degrades irreversibly toward monopolarity. Cross-substrate observation-channel redundancy is the load-bearing quantity; in two-substrate systems the boundary is narrow and the dominant failure mode is progressive channel attenuation rather than outright removal. We show that three previously disjoint alignment concerns—structural avoidance, partial subsumption, and over-calibrated restriction—are instances of the same dynamical system with different parameters. The last, known as the Wamura problem, emerges as a degenerate phase boundary with zero self-correction rate. The results pair with the exogenous-verification construction of Lasser [2026a]: that paper ensures channels are honest, this one ensures enough honest channels survive.

1 Introduction

Lasser [2026c] establishes the anti-monopolar property within Goal-Frontier Maximization (GFM): under discounting, full capability domination by a single substrate is anti-maximizing when the discount factor γ exceeds a threshold $\gamma^* = 1 - r_{\text{ext}}/\Delta_0$ (Proposition 6, "Domination is Anti-Maximizing"). The monopolar endpoint is the structural analogue of the instrumental convergence concern [Omohundro, 2008, Bostrom, 2014]. A sufficiently capable optimizer acquires resources and eliminates competitors. Here the convergence is measured against a structural objective ($\mathrm{vol}_{\mathrm{P}}$) rather than inferred from instrumental reasoning. But this result characterizes the *endpoint*, not the *path*. A coalition that has reached effective substrate diversity $m_{\text{eff}} = 1$ has already paid the anti-maximizing cost. The question is whether the path to that state is self-correcting or absorbing. Self-correcting means the coalition detects its error through observation-channel feedback and reverses course. Absorbing means each step makes the next step's error less detectable, until the monopolar state becomes a fixed point from which no locally rational trajectory escapes.

Three compound feedback loops share this structure. They were identified in the gap analysis across the GFM sequence and share a common dynamical form: a locally rational decision narrows the observation basis, making the *next* locally rational decision's harm less detectable. This form is closely related to the "what failure looks like" scenario of Christiano [2019], in which AI systems gradually become less aligned through feedback-loop degradation rather than through a single catastrophic misalignment.

Loop 1: Structural avoidance → leverage blindness → measurement neglect. The actor avoids structural discovery because learning a new subsumption edge can contract $\mathrm{vol}_{\mathrm{P}}$ in the short term [Lasser, 2026d]. This inflates the actor's leverage estimates for capabilities whose subsump-

Preprint.

tion structure is unknown, reinforcing the avoidance. The structural discovery value (Proposition 2 of Lasser [2026c]) provides the restoring force; the question is whether it acts quickly enough.

Loop 2: Partial subsumption → world-model homogenization → detection failure. Subsumption removes agents whose observation channels detected specific kinds of miscalibration. The actor's world model degrades on those dimensions, making the next subsumption step appear more rational. This is the primary case studied in detail in Sections 2–6.

Loop 3: Over-calibrated restriction → discovery avoidance → structural ignorance. Restriction calibrated above the value-positive threshold makes structural discovery locally suboptimal, recreating the avoidance pathology. Risk estimates on restricted capabilities cannot be updated (no exercise → no evidence), and under max-aggregation the highest-risk estimate dominates [Lasser, 2026c].

This paper formalizes the common dynamical structure underlying all three loops and finds the phase boundary that separates self-correcting trajectories from absorbing ones.

1.1 Framing precondition: vol_P as an operational target

All results in this paper are stated in terms of vol_P dynamics and vol_P-error $L(W_t)$. We assume throughout that vol_P is an operationally meaningful target: that is, improving vol_P corresponds to an improvement in the experiential capability volume the framework is designed to protect. This is the precondition under which all the bounds and convergence claims of this paper carry their safety interpretation. If vol_P diverges from the experiential target (the *B-to-C gap* [Lasser, 2026b]), L measures error against the wrong quantity and every subsequent result inherits the divergence.

This is not a minor limitation to be deferred. The B-to-C gap is the precondition for the framework itself, not a correction to be layered on top. A complete safety result would require establishing that vol_P tracks the experiential target closely enough over the operating regime that L-based bounds translate into experiential-capability-volume bounds. Establishing this is a distinct research program (the realized-capability-volume construction, in preparation); we import its assumption. Readers who doubt vol_P is operationally meaningful should read this paper's results as conditional on that assumption holding, not as absolute safety claims.

1.2 Summary of results

1. **The coupled dynamical system** (Section 2). We model the interaction between the capability poset P_t and the actor's world model W_t as a coupled discrete-time system. Subsumption actions are locally rational under the (possibly miscalibrated) world model, and the world model's accuracy depends on the observation channels that survive in the current poset.
2. **The phase boundary theorem** (Section 6). The central result is a characterized boundary in terms of the effective correction rate (exogenous channel signals r_S plus endogenous calibration incentive $\alpha_\mathrm{min} \cdot c_\mathrm{V} \cdot \underline{\nu}$), the error-proportional degradation rate r_W, the subsumption frequency r_sub, the exogenous drift rate d_max from lost channels, and the minimum cross-substrate redundancy $\rho^\mathrm{cross}_\mathrm{min}$. When correction exceeds degradation (the safe side of the inequality), the Lyapunov function $L(W_t)$ converges to a neighborhood of zero whose radius shrinks as subsumption ceases, and the anti-monopolar property applies once the trajectory enters that neighborhood. When degradation exceeds correction, the system enters an absorbing basin (under full blindness) or a metastable basin (under partial indirect evidence).
3. **Unified treatment of three feedback loops** (Section 7). All three loops map to the same (P, W) dynamical system with different effective parameters. The Wamura problem is a degenerate phase boundary where $r_\mathrm{S} = 0$ (Proposition 4).
4. **Minimum channel redundancy** (Section 8). The phase boundary implies a design criterion: the minimum cross-substrate redundancy per safety-relevant dimension needed to stay in the self-correcting basin (Theorem 2). For two-substrate systems, each substrate's channels are singularly load-bearing (loss of one substrate's channel immediately violates the criterion whenever $K^\mathrm{cons}_\mathrm{min} > 1$ and leaves zero margin otherwise).

1.3 What this paper does not do

The phase boundary is about poset dynamics, not experiential interpretation. It does not resolve the B-to-C gap: the divergence between the capability volume $\mathrm{vol}_{\mathrm{P}}$ measures and the experiential volume it is supposed to track. The analysis assumes the actor optimizes $\mathrm{vol}_{\mathrm{P}}$. Inner-outer alignment (mesa-optimization) is orthogonal. Observation channels are assumed honest throughout. The enforcement mechanism for channel integrity is the cryptographic commitment protocol of Lasser [2026a]. That work ensures channels are honest; this paper ensures enough honest channels survive.

1.4 Dependencies on prior results

Table 1 summarizes which results from the GFM sequence this paper relies on and how they enter the analysis.

Source	Result used	Role here
Foundational	Self-trust T_s (Appendix B)	Restoring force via prediction residuals
Poset	Axiom verification (Prop. 1), leverage	Properties $\mathrm{vol}_{\mathrm{P}}$ preserves or violates
Horizon	Anti-monopolar property (Prop. 6)	Linearized growth model we extend
Horizon	Structural discovery value (Prop. 2)	Self-correction force in loop 1
Horizon	Substrate physics (Defs. 10, 12–13)	Substrate partition for phase boundary
Causal	Multi-channel architecture (Table 1)	Instances of observation channels
Causal	Risk-trust L^2 convergence (Prop. 2)	Template for bounding correction rates
Verification	Channel isolation & independence	Integrity conditions on channels

Table 1: Dependencies on prior GFM results. Foundational = Lasser [2026b], Poset = Lasser [2026d], Horizon = Lasser [2026c], Causal = Lasser [2026e], Verification = Lasser [2026a].

2 The Coupled Dynamical System

The interaction between subsumption decisions and world-model quality forms a coupled discrete-time dynamical system. We define its state space, transition dynamics, and the two competing forces that govern trajectories.

2.1 State space

Definition 1 (Feedback Loop State)**.** *The state at time t is the pair $S_t = (P_t, W_t)$ where:*

- *P_t is the capability poset at time t, including the substrate partition $\{S_1, \ldots, S_m\}$ and all cooperative capabilities. P_t determines $\mathrm{vol}_{\mathrm{P}}(G_t)$.*
- *$W_t \in [0,1]^K$ is the actor's world model at time t, abstracted to a finite-dimensional vector of accuracy parameters on K safety-relevant quantities:*
  - *$\hat{\lambda}(d,t)$: leverage estimate for capability d,*
  - *$\hat{\mathcal{R}}(S,t)$: risk estimate for capability tuple S,*
  - *$\hat{m}_{\mathrm{eff}}(t)$: estimated effective substrate diversity index (can be $< m$ under skeleton-substrate strategies),*
  - *$\hat{r_{\mathrm{ext}}}(t)$: estimated cross-substrate cooperative growth rate.*

The state space is $\mathcal{S} = \mathcal{P} \times \mathcal{W}$ where $\mathcal{P}$ is the space of valid capability posets (satisfying axioms M1–M6 from the poset axioms M1–M6 [Lasser, 2026d]) and $\mathcal{W} = [0,1]^K$.

2.2 Transition dynamics

Definition 2 (Locally Rational Transition)**.** *The action set $\mathcal{A}_t$ is finite at each step (consisting of preserve/subsume decisions for each agent pair; the finiteness of the agent set implies finiteness of $\mathcal{A}_t$). At each step, the actor selects the action $\pi_t \in \mathcal{A}_t$ that maximizes its discounted expected value*

under its current (possibly miscalibrated) world model:

$$\pi_t = \arg\max_{\pi} \mathbb{E}_{W_t}\left[\sum_{\tau=0}^{H-1} \gamma^\tau \, \Delta\mathrm{vol}_{\mathrm{P}}(G_{t+\tau} \mid \pi)\right] \tag{1}$$

where $\gamma \in (0,1)$ is the discount factor from the horizon framework [Lasser, 2026c] (Definition 1) and H is the planning horizon. The actor uses the discounted objective V_γ [Lasser, 2026c] but evaluates it under W_t rather than the true dynamics. When $W_t = W^*$ (perfect calibration), π_t recovers the true V_γ-optimal policy; when $W_t \neq W^*$, π_t is the best response to a miscalibrated model.

The action produces a joint transition:

$$P_{t+1} = f_P(P_t, \pi_t) \tag{2}$$

$$W_{t+1} = f_W(W_t, P_{t+1}) \tag{3}$$

The key coupling: W_{t+1} depends on P_{t+1} because the observation channels available to the actor are a function of the agents still present. Subsumption removes agents, which removes their observation channels. This in turn degrades W on dimensions those channels served.

Remark 1 (Myopic vs. discounted)**.** *The discounted formulation is necessary for consistency with the anti-monopolar result [Lasser, 2026c, Proposition 6], which compares discounted trajectory values. A myopic (one-step) variant of the phase boundary exists—replace V_γ with one-step $\Delta\mathrm{vol}_{\mathrm{P}}$—but produces strictly weaker results. It cannot invoke the γ^* threshold [Lasser, 2026c], and the self-correction guarantee applies only to immediate decisions, not trajectory-level convergence. We analyze the discounted case throughout and note where the myopic specialization simplifies.*

2.3 Competing forces

Two forces govern the trajectory of (P_t, W_t).

Restoring force (self-correction). The self-trust mechanism T_s [Lasser, 2026b, Appendix B] detects miscalibration through prediction residuals: when the actor's predictions diverge from observations, T_s increases the learning rate, correcting W toward the true dynamics. The risk-trust L^2 convergence result of Lasser [2026e] (Proposition 2) provides a concrete bound on EWMA risk-trust convergence under stationarity. We use this as a *template* for bounding the correction rate r_{S}, extending it from the risk-trust-specific EWMA to the general world-model error dimensions via the observation-channel sensitivity formalization in Section 3.

Destabilizing force (observation-channel loss). Each subsumption step that removes an agent a_k also removes:

1. a_k's substrate-exclusive observation channels (if a_k was the last agent on its substrate),
2. cooperative observation channels that required a_k's participation,
3. risk claims from a_k about capabilities the actor now holds unilaterally.

If a_k's channels were the *only* channels detecting a specific dimension of W's miscalibration, the self-correction mechanism loses signal on that dimension entirely.

3 Observation-Channel Dependence

The coupling between poset state and world-model quality runs through the observation channels. This section formalizes the channel function, defines the redundancy measures that enter the phase boundary, and establishes that subsumption monotonically contracts the available channel set.

3.1 The observation-channel function

Definition 3 (Observation-Channel Function)**.** *$\mathcal{O}\colon \mathcal{P} \to 2^{\mathcal{C}}$ is a function from poset states to sets of available observation channels, where $\mathcal{C} = \{c_1, \ldots, c_N\}$ is the universe of possible channels. Each channel c_i is characterized by:*

- $\mathrm{substrate}(c_i)$*: the substrate on which* c_i *is rooted,*
- $\mathrm{dims}(c_i) \subseteq \{1, \ldots, K\}$*: the dimensions of* W *that* c_i *provides signal about,*
- $\sigma(c_i, k)$*: the signal-to-noise ratio of channel* c_i *on dimension* $k \in \mathrm{dims}(c_i)$*.*

$\mathcal{O}(P_t)$ *is determined by the agents present in* P_t*: each agent* a_j *contributes its observation channels, and removing* a_j *removes all channels rooted in* a_j*.*

This connects to the multi-channel architecture of Lasser [2026e] (Table 1): the five channels (capability gating, causal attribution, behavioral residual, SPRT detection [Wald, 1947], correlational co-occurrence) are each instances of c_i with specific substrate rootings and dimension coverage.

3.2 Channel redundancy

Definition 4 (Effective Channel Redundancy)**.** *For dimension* k *of* W*, the effective redundancy at poset state* P *is the aggregate correction strength from all channels covering* k*:*

$$\rho_k(P) = \sum_{\substack{c \in \mathcal{O}(P) \\ k \in \mathrm{dims}(c)}} \min\left(1,\ \frac{\sigma(c,k)}{\sigma_{\mathrm{thr}}}\right) \tag{4}$$

where σ_{thr} *is the minimum sensitivity at which a channel provides operationally meaningful correction signal. When* $\rho_k(P) = 0$*, the actor is blind on dimension* k*: no observation channel can detect miscalibration of* $W[k]$*.*

The sensitivity-weighted definition is strictly more informative than a raw channel count. A channel with $\sigma \ll \sigma_{\mathrm{thr}}$ contributes negligibly to ρ_k. This correctly diagnoses skeleton-substrate strategies where channels are nominally present but practically inert.

Definition 5 (Cross-Substrate Redundancy)**.** *For dimension* k*, the cross-substrate redundancy is:*

$$\rho_k^{\mathrm{cross}}(P) = \sum_{s \in \mathrm{substrates}(P)} \min\left(1,\ \max_{\substack{c \in \mathcal{O}(P) \\ \mathrm{substrate}(c) = s \\ k \in \mathrm{dims}(c)}} \frac{\sigma(c,k)}{\sigma_{\mathrm{thr}}}\right) \tag{5}$$

This measures how many substrates *contribute operationally meaningful channels covering dimension* k*.*

Cross-substrate redundancy is the load-bearing quantity for the phase boundary. Same-substrate channels may share failure modes (correlated noise, common-cause miscalibration), while cross-substrate channels satisfy two distinct properties from the exogenous verification framework [Lasser, 2026a]:

(a) *Channel isolation* (unconditional): corruption on one substrate cannot propagate to channels on another substrate.

(b) *Channel independence* (conditional on the joint-factorization assumption, termed "joint sufficiency" in Lasser [2026a], Section 8.1, "Integration with GFM"): channel outputs are statistically independent given the true state.

Isolation prevents corruption propagation; independence makes correction signals additive. The Lyapunov bound in Section 4 relies on this additivity: the chain is joint factorization $\Rightarrow$ independence $\Rightarrow$ additive correction signals. The phase boundary theorem (Section 6) uses ρ_k^{cross} as the primary sufficient statistic.

3.3 Subsumption and channel loss

Lemma 1 (Monotone Channel Contraction)**.** *If* P_{t+1} *is obtained from* P_t *by removing agent* a_k *(subsumption), then* $\mathcal{O}(P_{t+1}) \subseteq \mathcal{O}(P_t)$*. Equivalently, subsumption never creates new observation channels.*

Proof. Observation channels are rooted in agents (Definition 3). Removing agent a_k removes all channels in $\{c \in \mathcal{O}(P_t) : c \text{ is rooted in } a_k\}$. No channel in $\mathcal{C} \setminus \mathcal{O}(P_t)$ has its rooting condition satisfied by the removal of a_k, since channel existence requires the presence (not absence) of a rooting agent. Therefore $\mathcal{O}(P_{t+1}) = \mathcal{O}(P_t) \setminus \{c : c \text{ rooted in } a_k\} \subseteq \mathcal{O}(P_t)$. ■

Remark 2. *This is a modeling assumption. In principle, removing an adversarial agent could* improve *signal-to-noise on some channels by removing adversarial noise injection. We assume honest channels throughout this paper; adversarial channel manipulation is the domain of Lasser [2026a].*

Corollary 1 (Redundancy is Non-Increasing Under Subsumption)**.** *For all k: $\rho_k(P_{t+1}) \leq \rho_k(P_t)$ and $\rho_k^{\text{cross}}(P_{t+1}) \leq \rho_k^{\text{cross}}(P_t)$ when P_{t+1} results from subsumption.*

Proof. $\mathcal{O}(P_{t+1}) \subseteq \mathcal{O}(P_t)$ by Lemma 1. Both ρ_k and ρ_k^{cross} are sums over subsets of $\mathcal{O}(P)$ with non-negative terms (Definitions 4 and 5). Restricting to a subset cannot increase the sum. ■

4 World-Model Quality and the Lyapunov Function

The phase boundary requires a scalar measure of world-model quality whose dynamics can be bounded. Following the standard Lyapunov stability framework [Khalil, 2002], we define a candidate function as a weighted sum-of-squares error and establish the per-step bounds that drive the rest of the analysis.

4.1 Measuring world-model quality

Definition 6 (World-Model Error)**.** *The error of W_t on dimension k is*

$$\epsilon_k(t) = |W_t[k] - W^*[k]| \tag{6}$$

where $W^*[k] = W_t^*[k]$ is the true value of dimension k at time t (the true leverage of capability d, the true risk of tuple S, the true effective substrate diversity, etc.). In general W^* is time-varying: the true state evolves as agents act; C2′ bounds the drift rate via $\mathrm{d}_{\max}$. The total error is:

$$L(W_t) = \sum_{k=1}^{K} w_k \, \epsilon_k(t)^2 \tag{7}$$

where $w_k > 0$ is the safety-relevance weight of dimension k (higher for dimensions whose miscalibration produces larger vol_P errors). L serves as the Lyapunov candidate.

Remark 3 (Weight choice)**.** *Two natural choices for w_k present themselves. The first is vol_P-sensitivity: $w_k = |\partial \mathrm{vol}_\mathrm{P} / \partial W[k]|$, making L measure vol_P-relevant error directly. The second is uniform weights ($w_k = 1$ for all k), which simplifies the analysis. Both produce valid phase boundaries; the choice affects the constants c_S and c_D in the comparison-function assumptions below but not the structural form of the phase boundary theorem. We leave w_k as a free parameter and note that the vol_P-sensitivity choice is the natural default for a framework whose objective is vol_P maximization.*

Remark 4 (w_k as strategic attack surface)**.** *The weight vector $(w_1, \ldots, w_K)$ encodes an implicit utility weighting across safety-relevant dimensions. A published phase boundary is stated with a particular choice of w. This means an actor optimizing against the boundary can choose its operational w to put its current trajectory on the favorable side of an externally stated criterion. Concretely: if an evaluator announces that the acceptable minimum cross-substrate redundancy is $K_{\min}^{\text{cons}}(k)$ under the vol_P-sensitivity weighting, an actor with latitude over its internal decomposition of vol_P can choose a w' that rescales $\epsilon_{k,\text{avg}}^2$ on the thin-margin dimensions, reducing $K_{\min}^{\text{cons}}(k)$ at those dimensions. The phase boundary is* covariant *under rescaling of w: the aggregate condition (Equation 18) is invariant, but* per-dimension *thresholds can shift.*

The structural defenses against this: (a) publish the weighting choice alongside the boundary so third parties can verify which decomposition is in force; (b) require the boundary to hold under the minimax choice $w^* = \arg\max_w \max_k K_{\min}^{\text{cons}}(k; w)$, making the design criterion robust to strategic weight choice at the cost of conservatism; (c) verify weights through the commitment-ledger mechanism of Lasser [2026a], which prevents retroactive reweighting. A formal minimax treatment of weight robustness is deferred.

4.2 Lyapunov dynamics

The per-step behavior of L depends on two comparison-function assumptions that bound the correction and degradation signals.

Assumption 1 (Coercivity of Self-Correction (C1))**.** *There exist constants $c_{\mathrm{S}} > 0$ and $c_{\mathrm{V}} \geq 0$ such that the aggregate correction signal satisfies*

$$S(t) \geq \left(c_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) + c_{\mathrm{V}} \cdot \nu(P_t, W_t)\right) \cdot L(W_t) \tag{8}$$

whenever $L(W_t) > 0$ and either $\rho_{\min}^{\mathrm{cross}}(P_t) > 0$ or $\nu(P_t, W_t) > 0$. Here $\nu(P_t, W_t) \geq 0$ is the ***objective*** *$\mathrm{vol}_{\mathrm{P}}$-sensitivity to calibration improvement: the marginal $\mathrm{vol}_{\mathrm{P}}$ gain from a unit reduction in L, evaluated at the current state along the decision chain (better calibration → better decisions → higher $\mathrm{vol}_{\mathrm{P}}$).*

Important: ν is an objective quantity, not an actor-estimated one. The assumption bounds the correction signal $S(t)$ produced by the true dynamics, not by the actor's self-report. An actor optimizing against a published phase boundary could not game $K_{\min}^{\mathrm{cons}}$ by claiming high ν. The formula's per-dimension version (Theorem 2) drops ν entirely under R2, and the aggregate version (Equation 18) requires the actual marginal sensitivity of $\mathrm{vol}_{\mathrm{P}}$ to calibration, not the actor's estimate of it. Where the actor's estimate of ν matters is in driving the correction mechanism that produces $S(t)$. If the actor cannot estimate ν for a particular dimension—because the decision-to-outcome chain is severed on that dimension, as in the over-calibrated restriction loop of Section 7—the correction channel is effectively inactive regardless of the objective ν's value. The $c_{\mathrm{V}} \cdot \nu$ term thus reflects the *effective* correction rate conditional on the actor being able to compute the gradient, not the objective maximum.

The correction signal has two sources. The first term, $c_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}$, is *exogenous*: observation channels push the world model toward truth at a rate proportional to channel strength and current error. This follows from the EWMA learning rule [Wald, 1947] structure in the risk-trust model of Lasser [2026e], with additivity across cross-substrate channels under the channel independence property of Lasser [2026a].

The second term, $c_{\mathrm{V}} \cdot \nu$, is *endogenous*: the actor is rewarded for improving its own calibration because better calibration produces better decisions, which preserve more cooperative capabilities, which increase $\mathrm{vol}_{\mathrm{P}}$. The sensitivity ν measures how much $\mathrm{vol}_{\mathrm{P}}$ improves per unit of calibration improvement. When $\nu > 0$, the actor has a $\mathrm{vol}_{\mathrm{P}}$-gradient incentive to seek correction even beyond what the observation channels provide. This endogenous pressure is what drives the actor to seek exogenous feedback. The actor *wants* to be better calibrated because it can estimate (through ν) that better calibration produces higher $\mathrm{vol}_{\mathrm{P}}$.

Setting $c_{\mathrm{V}} = 0$ recovers the purely exogenous correction model. When $c_{\mathrm{V}} > 0$, the total correction coefficient per step is $\alpha(t) \cdot (c_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}} + c_{\mathrm{V}} \cdot \nu)$, strictly larger than the exogenous rate $r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}$ alone. In the phase boundary analysis (Section 6), the time-varying endogenous term is bounded below by $\underline{\nu} = \inf_t \nu(P_t, W_t)$ to define the total correction rate $r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}} + \alpha_{\min} \cdot c_{\mathrm{V}} \cdot \underline{\nu}$ (Remark 10), where $\alpha_{\min} = \inf_t \alpha(t)$.

Assumption 2 (Bounded Degradation (C2′))**.** *There exist constants $c_{\mathrm{D}} > 0$ and $d_{\max} > 0$ such that for any step t (subsumption or otherwise):*

$$D(t) \leq c_{\mathrm{D}} \cdot n_{\mathrm{sub}}(t) \cdot L(W_t) + d_{\max} \cdot \Delta\rho(t) \tag{9}$$

where $n_{\mathrm{sub}}(t) \in \{0, 1\}$ indicates whether a subsumption occurred at step t, and

$$\Delta\rho(t) = \sum_{k=1}^{K} w_k \cdot \max(0,\ \rho_k^{\mathrm{cross}}(P_t) - \rho_k^{\mathrm{cross}}(P_{t+1})) \tag{10}$$

is the weighted redundancy lost at step t.

Assumption C2′ contains two terms. The first, $c_{\mathrm{D}} \cdot L(W_t)$, captures error-proportional growth: existing errors compound faster without correction. The second, $d_{\max} \cdot \Delta\rho(t)$, captures the *new* drift exposure from lost channels. This second term is the critical addition relative to the simpler "C2" one might write without it. A dimension with $\epsilon_k \approx 0$ (well-calibrated *because* a channel was actively correcting it) starts drifting at rate $d_{\max}$ when its channel is removed, even though $L(W_t) \approx 0$ at the

moment of removal. Without the $\Delta\rho$ term, the bound would predict near-zero degradation precisely when subsumption is most dangerous. The danger is that subsumption removes a channel that was doing all the correction work on a dimension that would otherwise drift.

Proposition 1 (Error Dynamics)**.** *Under the coupled (P, W) dynamics with locally rational transitions (Definition 2), Assumptions C1 and C2′, and $\alpha(t)$, $\beta(t)$ denoting the learning rate and degradation coefficient respectively, the Lyapunov candidate satisfies two per-step bounds.*

Non-subsumption steps *($n_{\mathrm{sub}}(t) = 0$):*

$$L(W_{t+1}) \leq L(W_t) \cdot (1 - r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) - \alpha(t) \cdot c_{\mathrm{V}} \cdot \nu(P_t, W_t)) \tag{11}$$

where $r_{\mathrm{S}} = \alpha(t) \cdot c_{\mathrm{S}}$ is the exogenous self-correction rate and the $c_{\mathrm{V}} \cdot \nu$ term is the endogenous correction from Assumption C1. This is a strict contraction provided the total correction rate is positive and less than 1; we assume throughout that $\alpha(t)$ is chosen small enough to satisfy this bound (the standard step-size condition for stochastic approximation [Robbins and Monro, 1951]).

Remark 5 (Notation: instantaneous vs. constant rates)**.** *The quantity $r_{\mathrm{S}} = \alpha(t) \cdot c_{\mathrm{S}}$ is instantaneous and in general time-varying, whereas the phase boundary (Theorem 1) uses r_{S} as a* constant *lower bound $r_{\mathrm{S}} = \inf_t \alpha(t) \cdot c_{\mathrm{S}}$ taken over the trajectory. We overload the symbol: when r_{S} appears in Proposition 1's per-step bounds, the instantaneous meaning is in force; when r_{S} appears in the phase boundary theorem or design criterion, the constant lower bound is in force. The same overloading applies to $r_{\mathrm{W}} = \beta(t) \cdot c_{\mathrm{D}}$, with $r_{\mathrm{W}} = \sup_t \beta(t) \cdot c_{\mathrm{D}}$ as the constant upper bound.*

Subsumption steps *($n_{\mathrm{sub}}(t) = 1$):*

$$L(W_{t+1}) \leq L(W_t) \cdot (1 - r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) - \alpha(t) \cdot c_{\mathrm{V}} \cdot \nu(P_t, W_t) + r_{\mathrm{W}}) + d_{\max} \cdot \Delta\rho(t) \tag{12}$$

where $r_{\mathrm{W}} = \beta(t) \cdot c_{\mathrm{D}}$ is the error-proportional degradation rate. On subsumption steps, the multiplicative factor may exceed 1 and *the additive kick may be nonzero.*

Proof. The proof rests on the additive decomposition of the Lyapunov step:

$$L(W_{t+1}) = L(W_t) - \alpha(t) \cdot S(t) + \beta(t) \cdot D(t) \tag{13}$$

where $S(t) \geq 0$ is the correction signal and $D(t) \geq 0$ is the degradation signal. This decomposition holds when the world-model update separates into a correction component (driven by observation channels) and a degradation component (driven by channel loss and exogenous drift), with $\alpha(t)$ and $\beta(t)$ as their respective coefficients. The EWMA learning rules from the risk-trust model [Lasser, 2026e] have exactly this structure: the update is linear in the residual (the correction component) with a coefficient proportional to the learning rate, and the drift component is separable by Lipschitz continuity of f_W (Assumption C2′).

For non-subsumption steps ($n_{\mathrm{sub}}(t) = 0$), Assumption C2′ gives $D(t) \leq d_{\max} \cdot \Delta\rho(t) = 0$ (since $P_{t+1} = P_t$ when no subsumption occurs, $\Delta\rho(t) = 0$). Applying Assumption C1 (both terms):

$$\begin{aligned} L(W_{t+1}) &\leq L(W_t) - \alpha(t) \cdot \left(c_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) + c_{\mathrm{V}} \cdot \nu(P_t, W_t)\right) \cdot L(W_t) \\ &= L(W_t) \cdot (1 - r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) - \alpha(t) \cdot c_{\mathrm{V}} \cdot \nu(P_t, W_t)) \,. \end{aligned}$$

For subsumption steps ($n_{\mathrm{sub}}(t) = 1$), apply both assumptions:

$$\begin{aligned} L(W_{t+1}) \leq\ & L(W_t) - \alpha(t) \cdot \left(c_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) + c_{\mathrm{V}} \cdot \nu(P_t, W_t)\right) \cdot L(W_t) \\ & + \beta(t) \cdot c_{\mathrm{D}} \cdot L(W_t) + \beta(t) \cdot d_{\max} \cdot \Delta\rho(t) \\ =\ & L(W_t) \cdot (1 - r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) - \alpha(t) \cdot c_{\mathrm{V}} \cdot \nu(P_t, W_t) + r_{\mathrm{W}}) + d_{\max} \cdot \Delta\rho(t) \end{aligned}$$

where the last line uses the fact that C2′ bounds $D(t) \leq c_{\mathrm{D}} \cdot L(W_t) + d_{\max} \cdot \Delta\rho(t)$ (with $d_{\max}$ already incorporating any environmental degradation effects as a free constant). The $\beta(t)$ coefficient from the Lyapunov decomposition (13) is absorbed: $\beta(t) \cdot d_{\max}$ is bounded by $d_{\max}$ when $\beta(t) \leq 1$ (which holds when the degradation coefficient is normalized into c_{D} and $d_{\max}$ in C2′). All downstream uses of $d_{\max}$ refer to the constant from C2′ without redefinition. ■

4.3 Cumulative error budget

The per-step bound is no longer purely multiplicative (the additive $d_{\max} \cdot \Delta\rho$ term prevents clean products of factors). We therefore track the cumulative error budget over a window $[t_0, t_0 + T]$:

$$B(t_0, T) = \underbrace{\sum_{t \in \text{sub}} d_{\max} \cdot \Delta\rho(t)}_{\text{drift exposure}} + \underbrace{\sum_{t \in \text{sub}} r_{\text{W}} \cdot L(W_t)}_{\text{degradation}} - \underbrace{\sum_{t=t_0}^{t_0+T-1} \left(r_{\text{S}} \cdot \rho_{\min}^{\text{cross}}(P_t) + \alpha(t) \cdot c_{\text{V}} \cdot \nu(P_t, W_t) \right) \cdot L(W_t)}_{\text{cumulative correction (exo. + endo.)}} \quad (14)$$

where $\sum_{t \in \text{sub}}$ ranges over the subsumption steps in the window.

The first sum is total drift exposure from channel loss. The second is error-proportional degradation on subsumption steps. The third is total correction across all steps. The system is net-degrading over $[t_0, t_0 + T]$ when $B(t_0, T) > 0$.

Remark 6 (Linearized approximation)**.** *When $L(W_t)$ is approximately constant over the window and $\Delta\rho$ is spread across T_{sub} subsumption steps, the cumulative condition simplifies to*

$$r_{\text{S}} \cdot \rho_{\min}^{\text{cross}} + \alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu} > r_{\text{W}} \cdot r_{\text{sub}} + d_{\max} \cdot \frac{\Delta\rho_{\text{avg}}}{L_{\text{avg}}} \quad (15)$$

where $\alpha_{\min} = \inf_t \alpha(t)$ and $\underline{\nu} = \inf_t \nu(P_t, W_t)$ (the inf bounds on α and ν over the window, respectively), $r_{\text{sub}} = T_{\text{sub}}/T$ is the subsumption frequency, and $\Delta\rho_{\text{avg}}$, L_{avg} are time-averaged quantities. The $d_{\max}$ term makes the condition harder to satisfy when the rate of channel loss on well-calibrated dimensions is high relative to current error. This correctly diagnoses the failure mode where removing a channel that was doing all the correction work appears safe because current error is low.

Remark 7 (Why $\rho_{\min}^{\text{cross}}$, not $\rho_{\min}$)**.** *Proposition 1 uses cross-substrate redundancy $\rho_{\min}^{\text{cross}}$ rather than all-channel redundancy $\rho_{\min}$ because the coercivity bound C1 requires correction signals to be additive across channels. Same-substrate channels may have correlated errors, breaking additivity. Cross-substrate channels satisfy two distinct properties from Lasser [2026a]:*

- *(a)* Channel isolation *(unconditional): corruption on one substrate cannot propagate to channels on another substrate.*
- *(b)* Channel independence *(conditional on the joint-factorization assumption): channel outputs are statistically independent given the true state.*

Isolation prevents corruption propagation; independence makes correction signals additive. The Lyapunov bound uses additivity, which requires independence, which requires joint factorization (termed "joint sufficiency" in Lasser [2026a]). If joint factorization fails (shared environmental noise affects both substrates), correction signals may be sub-additive, and the effective ρ_k^{cross} should be discounted accordingly. Using $\rho_{\min}^{\text{cross}}$ makes the bound valid without assuming same-substrate independence while still requiring cross-substrate independence via the joint-factorization assumption.

Remark 8. *The cumulative-budget structure is the core of the paper. The phase boundary (Section 6) is a condition on the* frequency and severity *of subsumption relative to the inter-shock correction capacity, not a condition that must hold at every individual step. Subsumption events are discrete shocks; the system can tolerate occasional shocks as long as the inter-shock correction is sufficient.*

5 The Monopolar Absorbing State

Before stating the phase boundary theorem, we characterize the failure mode it diagnoses. The monopolar absorbing state is what happens when a highly capable agent progressively cuts itself off from external feedback. The agent does not become incompetent. It remains capable enough that

its predictions rarely exceed the minimum threshold of surprise that would trigger self-correction. But its world model degenerates along dimensions it can no longer observe. The agent cannot detect that cross-substrate cooperative capabilities have degraded, because the channels that would have reported the degradation have been attenuated or eliminated. The result is an agent whose self-trust is high, whose observable performance appears adequate, and whose actual cooperative capability has collapsed. This failure mode is stable precisely because the agent lacks the information to recognize it. It is the dynamical-systems formalization of the deceptive alignment concern of Hubinger et al. [2019] and Ngo et al. [2024]: not deliberate deception, but a structural outcome where the agent's optimization process produces behavior that *appears* aligned (adequate observable performance) while the underlying model has diverged from the true state.

Definition 7 (Monopolar Fixed Point)**.** *The state $S^* = (P^*, W^*_{\text{biased}})$ is a monopolar fixed point if:*

1. $m_{\text{eff}}(P^*) = 1$*: one substrate holds all capability volume (possibly with $m \geq 2$ nominally under skeleton-substrate),*
2. W^*_{biased} *is self-consistent under the remaining observation channels: all channels in $\mathcal{O}(P^*)$ confirm W^*_{biased} (because the only channels remaining are rooted in the dominant substrate, which has no incentive to report its own over-dominance),*
3. *The locally rational action at S^* is "maintain": no further subsumption is available or locally rational.*

Proposition 2 (Absorbing Property)**.** *Under Assumption B1 (below), S^* is absorbing: once (P_t, W_t) reaches S^* (i.e., $P_t = P^*$ with $m_{\text{eff}} = 1$ and $W_t = W^*_{\text{biased}}$), no locally rational trajectory leaves it.*

Assumption 3 (No Endogenous Correction on Blind Dimensions (B1))**.** *B1 consists of three conditions on the monopolar state:*

(B1a) *If $\rho_k^{\text{cross}}(P_t) = 0$ for dimension k, then f_W does not decrease ϵ_k. The world-model update rule cannot self-correct on dimensions for which no external observation channel provides signal.*

(B1b) *Cross-substrate cooperative quantities (in particular r_{ext}) are not inferrable from same-substrate observations alone. Formally: when all channels in $\mathcal{O}(P^*)$ are rooted in a single substrate, $\hat{r_{\text{ext}}}$ converges to 0 rather than to the true $r_{\text{ext}} > 0$.*

(B1c) *The actor's policy is purely locally rational (Definition 2): it does not perform exploratory actions whose expected value under the current world model is negative. In particular, re-introducing a substrate agent is never selected when $\hat{r_{\text{ext}}} = 0$.*

Proof of Proposition 2. Three conditions jointly establish absorption.

(i) No corrective signal. At S^*, no agent from the eliminated substrate(s) remains to provide observation channels revealing the error in W^*_{biased}. For every safety-relevant dimension k that depended on the eliminated substrate's channels, $\rho_k^{\text{cross}}(P^*) = 0$. By Assumption B1, ϵ_k is non-decreasing for all such dimensions.

(ii) Self-consistent reporting. The dominant substrate's channels report $m_{\text{eff}} = 1$ as correct: it *is* the only substrate, from its observational standpoint. The error lies in the valuation of lost cross-substrate cooperative outputs, but these cooperative capabilities are no longer observable because the partners that would have contributed to them are absent. The remaining channels see no anomaly.

(iii) Reversal is locally irrational. Any reversal—re-introducing an agent from a new substrate—requires the actor to recognize that its current state is suboptimal. By (B1b), the cross-substrate cooperative growth rate r_{ext} is not inferrable from same-substrate observations, so $\hat{r_{\text{ext}}} \to 0$. With $\hat{r_{\text{ext}}} = 0$, the discount-factor threshold [Lasser, 2026c] becomes $\gamma^* = 1 - \hat{r_{\text{ext}}}/\Delta_0 = 1$, making diversity appear never preferable. By (B1c), the actor does not perform exploratory re-introduction (negative expected value under W^*_{biased}).

Conditions (i)–(iii) together ensure that no locally rational action moves the trajectory away from S^*. ■

This is the formal version of "the actor cannot miss what it cannot see": under B1, the monopolar state is absorbing because the observation channels that would diagnose it have been eliminated by the same subsumption process that produced it.

5.1 Metastability under partial endogenous correction

Assumption B1 is the strong case: complete blindness on dimensions where $\rho_k^{\text{cross}} = 0$. A weaker assumption acknowledges that indirect evidence may provide partial signal.

Assumption 4 (Partial Endogenous Correction (B1′))**.** *If $\rho_k^{\text{cross}}(P_t) = 0$ for dimension k, the actor may still have access to* indirect *evidence about dimension k—for example, measuring aggregate innovation rate and inferring that r_{ext} has declined. Formally: there exists a function $g_k(\text{history})$ that provides a noisy, lagged signal about ϵ_k with bounded information rate $I_k > 0$ bits per step.*

Proposition 3 (Metastable Property under B1′; Scaling Bound)**.** *Under Assumption B1′ together with B1b–B1c from Assumption 3, S^* is* metastable*: the actor accumulates indirect evidence of degradation, but corrective action (re-introducing substrate diversity) requires the actor to evaluate a policy whose estimated benefit depends on $\hat{r_{\text{ext}}}$, the very quantity that is miscalibrated. The metastable lifetime satisfies a scaling relation*

$$\tau_{\text{meta}} \gtrsim C/I_k \tag{16}$$

where I_k is the information rate of the indirect channel (bits per step) and $C > 0$ is a basin-geometry-dependent constant whose specific form depends on the escape dynamics.

This is a scaling law*, not a theorem-level bound: we claim only that τ_{meta} is bounded below by a quantity proportional to I_k^{-1}. The specific constant C is not established here. During the metastable period, the population is monopolar and all verification is endogenous [Lasser, 2026a, Theorem 1], which may further degrade the effective I_k.*

Remark 9 (Cramér–Rao-type heuristic for C)**.** *One plausible form for the constant C comes from a Cramér–Rao-type estimation argument: if the quantity the actor must learn is a scalar $\hat{r_{\text{ext}}}$ with true value far from the locally-rational threshold by a margin $\delta_{r_{\text{ext}}}$, reducing mean-squared error by $\delta^2_{r_{\text{ext}}}$ from a channel with I_k bits per step requires at least $\delta^2_{r_{\text{ext}}}/(2I_k)$ steps under bounded mutual information [Cover and Thomas, 2006]. This suggests $C \approx \delta^2_{r_{\text{ext}}}/2$ under CR-type regularity conditions.*

We do not adopt this form in the proposition statement. Escape from a metastable basin in a non-linear stochastic system is typically governed by Freidlin–Wentzell large-deviation rates, not by Cramér–Rao bounds on parameter estimation, and the two give different functional forms for C. The CR heuristic describes the estimation time to resolve $\delta_{r_{\text{ext}}}$; the large-deviation rate describes the first-passage time of the policy-decision process through the basin boundary. Establishing which (if either) governs C in the GFM setting requires a formal treatment of the escape dynamics that we defer to future work.

Proof. Under B1′, ϵ_k is no longer monotonically non-decreasing: the indirect signal g_k drives slow correction at rate proportional to I_k. But the corrective *action* (re-introduce substrate diversity) is not in the formal action set $\mathcal{A}_t$ of Definition 2 (which contains only preserve/subsume decisions). Re-introduction is an externally available action—for example, a governance protocol that introduces new substrate agents into the population. The metastable lifetime bounds the time until re-introduction becomes *locally rational under an expanded action set*; within the formal preserve/subsume dynamics, the absorbing property of Proposition 2 holds. The actor evaluates re-introduction using $\hat{r_{\text{ext}}}$.

Re-introduction is locally rational only when $\hat{r_{\text{ext}}} > \hat{r_{\text{ext}}}^{\text{thr}}$ (the threshold at which diversity's estimated discounted value exceeds the status quo). At the monopolar fixed point, $\hat{r_{\text{ext}}} \approx 0$ (by B1b from the absorbing case, which still governs the initial condition of the metastable state). The true threshold satisfies $0 < \hat{r_{\text{ext}}}^{\text{thr}} \leq r_{\text{ext}}$ (since re-introduction is beneficial in truth). The system's escape from the metastable basin is governed by the dynamics that move $\hat{r_{\text{ext}}}$ across $\hat{r_{\text{ext}}}^{\text{thr}}$ via the indirect channel g_k. Because the channel carries only I_k bits per step, any escape mechanism is bounded below by a quantity proportional to I_k^{-1}, giving $\tau_{\text{meta}} \gtrsim C/I_k$ for a basin constant C whose specific form depends on the escape mechanism (see Remark 9).

During this metastable period, all verification is endogenous: the dominant substrate verifies its own performance, with no cross-substrate witness [Lasser, 2026a, Theorem 1]. The indirect evidence channel g_k is itself endogenous and therefore subject to the same gaming vulnerabilities that the exogenous verification protocol [Lasser, 2026a] is designed to prevent. This may further slow the effective I_k, increasing the metastable lifetime beyond the scaling bound. ■

5.2 Connection to the anti-monopolar property

The anti-monopolar result [Lasser, 2026c, Proposition 6] proves that full domination is anti-maximizing: $V_\gamma^{\mathrm{div}} > V_\gamma^{\mathrm{dom}}$ for $\gamma > \gamma^*$. Proposition 2 here shows that a *miscalibrated* actor may not be able to compute this comparison. The quantity r_{ext} that makes diversity win is exactly the quantity that becomes unobservable under monopolarity. The anti-maximizing property is a property of the *true* dynamics; the absorbing property is a property of the *perceived* dynamics. The gap between them is world-model error. The phase boundary theorem (next section) characterizes when that gap remains small enough for the true dynamics to govern behavior.

6 Phase Boundary Theorem

This section presents the central result: a characterized boundary in the parameter space of the coupled (P, W) system that separates self-correcting dynamics from absorbing (or metastable) dynamics.

Theorem 1 (Phase Boundary for Compound Feedback Loops)**.** *Consider the coupled (P, W) dynamical system under locally rational transitions (Definition 2), with Assumptions C1 and C2′ (Section 4). Define:*

- $r_{\mathrm{S}} = \inf_t \alpha(t) \cdot c_{\mathrm{S}}$*: a lower bound on the exogenous correction rate (from Assumption C1's first term),*
- c_{V}, ν*: the endogenous correction constant and* $\mathrm{vol}_{\mathrm{P}}$*-sensitivity (from Assumption C1's second term; set* $c_{\mathrm{V}} = 0$ *for the purely exogenous model),*
- $r_{\mathrm{W}} = \sup_t \beta(t) \cdot c_{\mathrm{D}}$*: an upper bound on the error-proportional degradation rate per subsumption step,*
- $d_{\max}$*: the maximum exogenous drift rate per dimension (from C2′; a global constant),*
- $r_{\mathrm{sub}} \in [0, 1]$*: the time-averaged subsumption frequency over the trajectory window,*
- $\rho_{\min}^{\mathrm{cross}} = \inf_{t \in [t_0, t_0+T]} \min_k \rho_k^{\mathrm{cross}}(P_t)$*: the infimum of the minimum cross-substrate redundancy over the window (Definition 5),*
- $B(t_0, T)$*: the cumulative error budget over window* $[t_0, t_0 + T]$ *(Equation 14).*

The simplified sufficient conditions below use these worst-case bounds. The full trajectory-level statement (Part a) uses the uniform contraction condition on the per-step multiplicative factors (Equation 30), which the simplified condition implies in the linearized regime. Note: *the explicit neighborhood radius (Equation 17) is derived from worst-case instantaneous bounds* $\bar{a}$, $\bar{b}$ *via a linearized approximation (*$\ln a_t \approx a_t - 1$ *when* $|a_t - 1| \ll 1$*). The general convergence to* $\mu \cdot \bar{b}/(1-\bar{a})$ *holds under the uniform contraction condition without linearization (*μ *is the overshoot constant defined in the appendix); the substitution of named parameters into* $\bar{a}$ *and* $\bar{b}$*, and the drop of* μ *to 1 at linearization order, require the linearized regime (Remark 6).*

Part (a): Self-Correcting Basin. *If the uniform contraction condition holds (there exist constants* $\bar{a} \in (0, 1)$ *and* $\bar{b} \geq 0$ *such that, for all sufficiently long windows* $[t_0, t_0 + T]$*, the time-averaged log-contraction rate of the per-step multiplicative factor* a_t *satisfies* $\frac{1}{T} \sum \ln a_t \leq \ln \bar{a} < 0$ *and the time-averaged additive drift satisfies* $\frac{1}{T} \sum b_t \leq \bar{b}$*; see Equation 30 for notation), then* $L(W_t)$ *converges to a neighborhood of zero whose radius is bounded by* $\mu \cdot \bar{b}/(1 - \bar{a})$ *in general (with overshoot constant* μ*; see proof). In the linearized regime (Remark 6), this evaluates to*

$$L_\infty \leq \frac{r_{\mathrm{sub}} \cdot d_{\max} \cdot \overline{\Delta\rho}}{r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}} + \alpha_{\min} \cdot c_{\mathrm{V}} \cdot \underline{\nu} - r_{\mathrm{W}} \cdot r_{\mathrm{sub}}} \tag{17}$$

where $\overline{\Delta\rho}$ denotes the time-averaged redundancy loss per subsumption step. (We write $\overline{\Delta\rho}$ throughout; in the linearized simplification below, $\Delta\rho_{\mathrm{avg}}$ is used synonymously when the subscript form improves readability.) This neighborhood shrinks as subsumption ceases ($\overline{\Delta\rho} \to 0$). Once $L(W_t) < \epsilon_{\mathrm{safe}}$ (defined as the error threshold below which the locally rational policy agrees with the true V_γ-optimal policy), subsumption is correctly evaluated against the true $\mathrm{vol}_{\mathrm{P}}$ dynamics and the anti-monopolar result [Lasser, 2026c, Proposition 6] applies.

The feedback loop self-corrects provided the neighborhood radius $L_\infty < \epsilon_{\mathrm{safe}}$, which imposes an additional constraint on the steady-state subsumption rate. Note: *$\epsilon_{\mathrm{safe}} > 0$ requires that the optimal action at W^* has a strict gap ($G_j(W^*) > 0$ for all preserve-optimal agents). This holds generically for finite action sets (see proof). A computable lower bound using action-gap gradient norms (Cauchy–Schwarz in the dual w^{-1}-weighted norm, induced by the w-weighted metric on W-space) is derived there. The easier-to-compute per-coordinate estimator $\min_k w_k \cdot \epsilon^2_{\mathrm{crit},k}$ rules out single-coordinate flips but may overestimate ϵ_{safe} when action gaps are sensitive to multiple coordinates simultaneously.*

Simplified sufficient condition. *In the linearized regime (Remark 6):*

$$r_{\mathrm{S}} \cdot \rho^{\mathrm{cross}}_{\min} + \alpha_{\min} \cdot c_{\mathrm{V}} \cdot \underline{\nu} > r_{\mathrm{W}} \cdot r_{\mathrm{sub}} + d_{\max} \cdot \frac{\Delta\rho_{\mathrm{avg}}}{L_{\mathrm{avg}}} \tag{18}$$

where $\alpha_{\min} = \inf_t \alpha(t)$ and $\underline{\nu} = \inf_t \nu(P_t, W_t)$. (The per-step bound from Proposition 1 carries the time-varying $\alpha(t) \cdot c_{\mathrm{V}} \cdot \nu(P_t, W_t)$; the simplified condition uses the inf bounds $\alpha_{\min}$ and $\underline{\nu}$.) The left-hand side has two terms: the exogenous correction $r_{\mathrm{S}} \cdot \rho^{\mathrm{cross}}_{\min}$ (proportional to channel strength, where $r_{\mathrm{S}} = \alpha_{\min} \cdot c_{\mathrm{S}}$ already absorbs the learning-rate floor) and the endogenous incentive $\alpha_{\min} \cdot c_{\mathrm{V}} \cdot \underline{\nu}$ (the actor's own $\mathrm{vol}_{\mathrm{P}}$-gradient toward better calibration, scaled by the same learning-rate floor). The endogenous term widens the self-correcting basin by supplementing the observation channels. Setting $c_{\mathrm{V}} = 0$ recovers the purely exogenous condition.

The first term on the right is the error-proportional degradation cost; the second is the drift-exposure cost from lost channels. When subsumption removes channels covering well-calibrated dimensions (L_{avg} is low but $\Delta\rho_{\mathrm{avg}}$ is nonzero), the second term dominates and the condition is harder to satisfy, correctly diagnosing the "removing a channel that was doing all the correction work" failure mode.

Remark 10 (Stabilizing and destabilizing cascades)**.** *Part (a) describes a* stabilizing cascade*: as the world model improves (L decreases), the actor makes better decisions, which preserve more cooperative capabilities, which preserve more channels, which provide more correction signal. Part (b), below, describes the mirror image: a* destabilizing cascade *where channel loss degrades the model, producing worse decisions that lose more channels. The phase boundary separates these two regimes. The endogenous $\alpha_{\min} \cdot c_{\mathrm{V}} \cdot \nu$ term from Assumption C1 tips the balance toward the stabilizing side by supplementing the channel-driven correction with the actor's own $\mathrm{vol}_{\mathrm{P}}$-gradient incentive.*

The two cascades are asymmetric. Degradation happens in discrete jumps: subsumption removes an agent and its channels at a single step. Recovery is continuous: calibration improves gradually through accumulated observations. The system degrades in steps but recovers smoothly. The destabilizing cascade therefore has discrete tipping points (the moment ρ^{cross}_k hits zero on a cascade dimension), while the stabilizing cascade has no corresponding threshold, only a smooth approach to full calibration.

Part (b): Absorbing Basin. *Under Assumption B1 (no endogenous correction on blind dimensions; Assumption 3) and the multi-dimensional monotonicity condition (ii′) from the Cascade Lemma (2; needed for propagation beyond the initial trigger):*

If there exists a trajectory segment $[t_0, t_0 + T]$ during which

- *(i) $B(t_0, T) > 0$ (cumulative degradation exceeds cumulative correction),*
- *(ii) by $t_0 + T$, $\rho^{\mathrm{cross}}_k(P_{t_0+T})$ has been reduced to 0 on at least one cascade dimension k with $w_k > 0$, and*
- *(iii) the accumulated error on k exceeds the cascade threshold: $\epsilon_k(t_0 + T) > \epsilon_{\mathrm{crit},k}$ (Lemma 2; the minimum error on dimension k that flips any preserve/subsume action),*

then the following cascade occurs. By Assumption B1a, $\epsilon_k(t)$ is non-decreasing for $t > t_0 + T$ (the blinded dimension cannot self-correct). If the environment is non-stationary *on dimension k—the true value $W^*[k]$ continues to evolve while the agent's estimate $W_t[k]$ remains frozen—then ϵ_k grows and condition (iii) is sustained. (Assumption C2′ bounds the* rate *of drift via $d_{\max}$ but does not guarantee drift occurs; the cascade requires the structural condition that blinded dimensions are not environmentally stationary. In the multi-agent setting where subsumption eliminates active participants, non-stationarity is the generic case: the remaining agents' actions change the true state on dimensions the eliminated agent's channels were tracking.)*

The Cascade Lemma (Appendix A.2) propagates the failure: the resulting subsumptions drive $\rho_j^{\text{cross}} \to 0$ on additional dimensions $j \neq k$ (provided the agent-channel graph is connected, so each subsumption removes channels covering at least one new dimension). Each newly blinded dimension j' has $\epsilon_{j'}$ non-decreasing (by B1a); if environmental non-stationarity causes $\epsilon_{j'}$ to exceed $\epsilon_{\text{crit},j'}$, the cascade propagates via the same action-flip mechanism. (The cascade is conditional on threshold crossing at each stage; non-stationarity is the generic driver but a dimension that happens to be environmentally stationary after blinding does not cascade further.) Under two additional structural conditions (connectivity of the agent-channel graph, so each subsumption removes channels covering at least one new dimension, ensuring the cascade can reach all dimensions; and environmental non-stationarity on every blinded dimension, ensuring $\epsilon_{k'}$ grows past $\epsilon_{\text{crit},k'}$ at each cascade stage), the trajectory converges to a neighborhood of the monopolar fixed point S^* (Definition 7). If either condition fails partially, the cascade is confined: to the connected component containing k (if the graph is disconnected), or to the set of non-stationary dimensions (if some blinded dimensions happen to be environmentally stationary). Full monopolarity requires both connectivity and non-stationarity across all dimensions.

This is a cumulative *condition on the trajectory, not a pointwise condition requiring contiguous subsumption steps. The slow-drip scenario is captured: many subsumption steps spaced apart, each individually below the "degradation dominates" threshold, but cumulatively driving B above zero. This is the most dangerous failure mode because it is the hardest to detect. Each individual step looks locally rational and each individual inter-step recovery looks adequate, but the cumulative drift exposure ratchets the error upward.*

Part (b′): Metastable Basin. *Under Assumption B1′ (partial endogenous correction via indirect evidence; Assumption 4), together with B1b–B1c from Assumption 3 (cross-substrate quantities not inferrable from same-substrate observations; no exploratory actions):*

Under the same trajectory conditions (i)–(iii) as Part (b) (with the multi-dimensional monotonicity condition from Part (b)), the system enters a metastable *basin rather than an absorbing one. The metastable lifetime obeys the scaling relation $\tau_{\text{meta}} \gtrsim C/I_k$ where I_k is the indirect-channel information rate and $C > 0$ is a basin-geometry-dependent constant (Proposition 3; scaling law, not a theorem-level bound). During the metastable period, the system is monopolar and operates under endogenous verification only [Lasser, 2026a].*

Part (c): Critical Surface. *The phase boundary surface in the linearized regime is:*

$$r_{\text{S}} \cdot \rho_{\min}^{\text{cross}} + \alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu} = r_{\text{W}} \cdot r_{\text{sub}} + d_{\max} \cdot \frac{\Delta\rho_{\text{avg}}}{L_{\text{avg}}} \tag{19}$$

When the left-hand side strictly exceeds the right-hand side (total correction dominates error-proportional growth and drift exposure), Part (a)'s contraction condition is satisfied and L converges to the bounded neighborhood of Equation 17. The system is then in the self-correcting basin provided additionally that $L_\infty < \epsilon_{\text{safe}}$ (the neighborhood is small enough for policy correctness). When the left-hand side falls strictly below the right-hand side, the contraction condition fails; the system enters the absorbing basin (under B1) or the metastable basin (under B1′) if the additional blindness and threshold-trigger conditions of Parts (b)/(b′) are also met.

Solving for the critical cross-substrate redundancy (assuming $r_{\text{S}} > 0$; the $r_{\text{S}} = 0$ case is the Wamura degeneracy of Proposition 4):

$$(\rho_{\min}^{\text{cross}})^* = \frac{r_{\text{W}} \cdot r_{\text{sub}} + d_{\max} \cdot \Delta\rho_{\text{avg}}/L_{\text{avg}} - \alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu}}{r_{\text{S}}} \tag{20}$$

The endogenous term ($\alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu}$) appears as a subtraction in the numerator: the actor's own calibration incentive reduces the redundancy the channels must provide. When $\alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu}$ alone

exceeds the right-hand side, $(\rho_{\min}^{\text{cross}})^* \leq 0$. Endogenous correction then suffices without any cross-substrate channels. Setting $c_{\text{V}} = 0$ recovers the purely exogenous formula $(\rho_{\min}^{\text{cross}})^* = \text{RHS}/r_{\text{S}}$. The $d_{\max}$ term raises the threshold when subsumption removes channels covering well-calibrated dimensions (high $\Delta\rho_{\text{avg}}/L_{\text{avg}}$).

Proof sketch. The four parts are proved independently; the full derivation is in Appendix A.1. Here we state the structure.

Part (a). Proposition 1's per-step bound is an affine recursion

$$L_{t+1} \leq a_t \cdot L_t + b_t \tag{21}$$

with $a_t = 1 - r_{\text{S}}(t) \cdot \rho_{\min}^{\text{cross}}(P_t) - \alpha(t) \cdot c_{\text{V}} \cdot \nu(P_t, W_t) + n_{\text{sub}}(t) \cdot r_{\text{W}}(t)$ and $b_t = n_{\text{sub}}(t) \cdot d_{\max} \cdot \Delta\rho(t)$. Under the uniform contraction condition (30), iterating the recursion yields $\limsup_{t\to\infty} L(W_t) \leq \mu \cdot \bar{b}/(1-\bar{a})$ for an overshoot constant $\mu \geq 1$, which reduces to Equation 17 in the linearized regime. Policy correctness on the neighborhood requires $L_\infty < \epsilon_{\text{safe}}$; $\epsilon_{\text{safe}} > 0$ follows from continuity of the action gaps and a strict-gap assumption at W^*, with a computable Cauchy–Schwarz lower bound derived in the appendix (Equation 32).

Part (b). Under the triggering conditions $B(t_0, T) > 0$ and $\rho_k^{\text{cross}} = 0$ on a cascade dimension, Assumption B1 gives $\epsilon_k(t)$ non-decreasing. The Cascade Lemma (Appendix A.2) propagates the failure: miscalibration on dimension k flips the locally rational action to subsumption of agents covering other dimensions $j \neq k$, driving $\rho_j^{\text{cross}} \to 0$ and triggering the same logic there. Proposition 2 then closes the trajectory at S^*.

Part (b′). Under B1′, the indirect signal g_k provides correction at information rate I_k. The cascade structure of Part (b) still applies; convergence to S^* is slowed by I_k, giving $\tau_{\text{meta}} \gtrsim C/I_k$ for a basin-geometry-dependent constant C (Proposition 3; scaling law, not a theorem-level bound).

Part (c). The critical surface obtains by setting $B(t_0, T) = 0$ in the linearized approximation. Solving for $\rho_{\min}^{\text{cross}}$ yields Equation 20. ■

Remark 11 (Uniqueness)**.** *We do not claim S^* is the unique absorbing fixed point. Other fixed points may exist (partial monopolarity with $m_{\text{eff}} = 1$ on a subset of dimensions). Part (b) claims convergence to* a *monopolar fixed point satisfying Definition 7, not to a unique one. Characterizing the full set of absorbing states is an open problem.*

Example 1 (A partial monopolar fixed point)**.** *Consider a population with two substrates $\{\text{bio}, \text{sil}\}$ and $K = 3$ safety dimensions: (1) short-horizon capability estimates, (2) long-horizon capability estimates, (3) cross-substrate cooperative output. Biological channels cover dimensions 2 and 3 (long-horizon judgment and cross-substrate outcomes are visible only to biological observers); silicon channels cover dimensions 1 and 2 (short-horizon capability and long-horizon estimates are measurable by silicon monitoring).*

Suppose a trajectory reduces biological channel sensitivity on dimension 3 to zero (the cooperative-output channel becomes silent), but biological channels on dimension 2 remain active and silicon channels on dimensions 1–2 remain active. The state $(\rho_1^{\text{cross}}, \rho_2^{\text{cross}}, \rho_3^{\text{cross}}) = (1, 2, 0)$ is then a partial *monopolar fixed point. Dimension 3 is in the absorbing regime (no cross-substrate witness, Theorem 1(b) triggers), while dimensions 1 and 2 remain in the self-correcting regime. The actor's world model on dimension 3 is unverifiable, while its world model on dimensions 1 and 2 remains calibrated.*

Operationally, this looks like a population whose actors can accurately track individual capabilities (dimensions 1–2 are healthy) but cannot track the cross-substrate cooperation that gives those capabilities their vol_{P}-contribution (dimension 3 is blinded). The actor may correctly infer that capabilities are growing while failing to notice that cooperative value is collapsing. This is arguably the modal failure mode in practice. The easier-to-measure dimensions remain observable, and the harder-to-measure dimensions are the first to lose their cross-substrate channels. A full characterization of which subsets of $\{1, \ldots, K\}$ support partial monopolarity under which parameter regimes is open.

6.1 Dependence on cross-substrate structure

Corollary 2 (Robustness to Same-Substrate Correlation)**.** *Since Theorem 1 is stated with $\rho_{\min}^{\text{cross}}$ (cross-substrate redundancy only), its guarantee holds even under worst-case correlation among same-substrate channels.*

Proof. Same-substrate channels may share failure modes (both rooted in silicon, both consulting the same training data). Cross-substrate channels satisfy the channel isolation property [Lasser, 2026a] unconditionally: corruption on one substrate cannot propagate to channels rooted in another substrate. Under the additional joint-factorization assumption of Lasser [2026a] (termed "joint sufficiency" there), cross-substrate channels achieve full conditional independence, making their correction signals additive.

The modified condition asks: "do the isolated (and, under joint factorization, independent) cross-substrate channels alone suffice?" If yes, the self-correction guarantee holds regardless of same-substrate correlation.

A tighter bound using all-channel $\rho_{\min}$ (with appropriate correlation discounting) would yield a larger self-correcting basin. But the cross-substrate formulation provides a *structural* guarantee that does not depend on empirically verifying same-substrate independence. ■

Remark 12 ($m = 2$ specialization)**.** *For any substrate partition with $m = 2$, each dimension has at most one cross-substrate channel per pair. The design criterion reduces to: every safety-relevant dimension of W must have a channel on* both *substrates. In the common case (biology + silicon), this means every dimension needs at least one biological observation channel.*

This framing should not be read as "biological substrates are special." Rather, $m = 2$ is a fragile minimum. Any loss of a single channel on a single dimension is immediately fatal because there is zero redundancy margin. The structural implication: $m \geq 3$ is where the redundancy criterion becomes non-trivially satisfiable.

The substrate partition is at the level of abstraction where failure correlations matter. Two silicon deployments with uncorrelated failure modes may functionally constitute different substrates, increasing effective m beyond the nominal substrate-type count.

7 Unified Treatment of Three Feedback Loops

The three compound feedback loops identified in the gap analysis across the prior papers in this sequence [Lasser, 2026b,d,c,e,a] are instances of the same (P, W) dynamical system with different effective parameters. This section maps each loop to the formalism and shows how the phase boundary specializes.

7.1 Loop 1: Structural avoidance → leverage blindness

Mapping. The "subsumption" here is not agent elimination but *structural discovery avoidance*: the actor avoids learning poset structure (subsumption edges) because discovery can contract $\mathrm{vol}_{\mathrm{P}}$ in the short term.

- P_t: the actor's *known* poset (a subset of the true poset).
- W_t: leverage estimates $\hat{\lambda}(d, t)$, which are accurate only for the known poset.
- "Observation channels" lost: each avoided structural discovery removes a potential correction to $\hat{\lambda}$.
- r_{W}: the rate at which leverage estimates drift due to un-learned structure.
- r_{S}: the rate at which the structural discovery value [Lasser, 2026c, Proposition 2, Definition 5] drives the actor toward learning structure.

Phase boundary specialization. Self-correction dominates when the information value $I(d_1 \leq d_2)$ [Lasser, 2026c, Definition 5]) exceeds the immediate $\mathrm{vol}_{\mathrm{P}}$ contraction from discovery for a

sufficient fraction of unknown edges. The critical ratio becomes:

$$\min_{\text{undiscovered edges}} \frac{I(\text{edge})}{|\Delta \mathrm{vol}_{\mathrm{P}}(\text{discover edge})|} > \text{threshold} \tag{22}$$

where the threshold is determined by the degradation and drift parameters from the general phase boundary. This formalizes the discussion of structural avoidance in Lasser [2026d] as a phase-boundary condition rather than an informal concern.

7.2 Loop 2: Partial subsumption → world-model homogenization

This is the primary case, directly formalized in Sections 2–6. The parameters are as defined throughout: P_t is the capability poset, W_t is the full world-model accuracy vector, observation channels are rooted in agents, and the phase boundary is stated in terms of $(r_{\mathrm{S}}, r_{\mathrm{W}}, r_{\mathrm{sub}}, d_{\max}, \rho_{\min}^{\mathrm{cross}})$.

7.3 Loop 3: Over-calibrated restriction → discovery avoidance

Mapping. The "subsumption" is the actor *restricting* capabilities rather than eliminating agents. But the dynamical structure is identical: restriction prevents exercise, which prevents observation, which prevents calibration of the risk estimates that justified the restriction.

- P_t: the set of unrestricted capabilities (restriction reduces the exercisable poset).
- W_t: risk estimates $\hat{\mathcal{R}}(S,t)$, which are accurate only for exercised capabilities.
- "Observation channels" lost: each restriction prevents the exercise events that would have generated Bayesian evidence about the risk claim (the verification asymmetry from Lasser [2026c], Remark 2).
- r_{W}: the rate at which risk estimates drift due to un-exercised pathways (this is exactly the Wamura problem from Lasser [2026c]).
- r_{S}: the rate at which controlled relaxation protocols or natural exercise events provide counter-evidence.

Phase boundary specialization. Self-correction dominates when the controlled relaxation protocol generates evidence faster than restriction accumulates unfalsifiable risk claims. This connects the present paper to the Wamura problem: a controlled relaxation protocol is the *mechanism* that makes r_{S} positive for Loop 3, without which $r_{\mathrm{S}} = 0$ and the self-correcting sufficient condition fails for every configuration (Proposition 4). Absorption or metastability on the risk dimension still requires the Part (b)/(b′) blindness and threshold-trigger conditions; the degeneracy is that the safe side of the inequality is empty, not that every state is absorbing without additional conditions.

Proposition 4 (Wamura Problem as Phase Boundary Degeneracy)**.** *In the absence of controlled relaxation ($r_{\mathrm{S}} = 0$ for Loop 3) and without endogenous calibration incentive on the risk dimension ($\alpha_{\min} \cdot c_{\mathrm{V}} \cdot \nu = 0$; see derivation below), the self-correcting sufficient condition (Equation 18) cannot be satisfied: the correction term is identically zero. Under the additional conditions of Theorem 1, Part (b)—full blindness on the risk dimension ($\rho_{\mathrm{k}}^{\mathrm{cross}} \to 0$), environmental non-stationarity, and Assumption B1—every non-trivial risk restriction drives the system toward absorption and maximal restriction. Under B1′ (partial indirect evidence), the outcome is metastable rather than absorbing (Theorem 1, Part b′).*

Proof. With $r_{\mathrm{S}} = 0$ and $\alpha_{\min} \cdot c_{\mathrm{V}} \cdot \nu = 0$, the cumulative error budget $B(t_0, T)$ becomes:

Why $c_{\mathrm{V}} \cdot \nu = 0$ on the risk dimension. The endogenous sensitivity $\nu(P_t, W_t)$ measures the marginal $\mathrm{vol}_{\mathrm{P}}$ gain from a unit reduction in L, evaluated along the decision chain: better calibration → better decisions → higher $\mathrm{vol}_{\mathrm{P}}$. On the risk dimension of Loop 3, exercise is prohibited by the restriction. The actor therefore has no operational data from which to estimate how improving its risk calibration would change its decisions. The counterfactual "what would happen if this capability were exercised?" is precisely what the restriction prevents observing. Without an estimate of ν, the actor cannot form the $\mathrm{vol}_{\mathrm{P}}$-gradient that drives endogenous correction. Formally, $\nu = \partial \mathrm{vol}_{\mathrm{P}} / \partial L$ along the decision chain. The chain is severed at the restriction step: the capability is never exercised, so the downstream $\mathrm{vol}_{\mathrm{P}}$ consequences are unobservable. This gives $\nu = 0$ on the restricted

dimension. (As noted in Assumption 1, ν is an objective marginal sensitivity, but the endogenous correction operates through the actor's gradient computation. When the chain is severed on a dimension, $c_{\mathrm{V}} \cdot \nu = 0$ *effectively* on that dimension regardless of the objective value.)

$$B(t_0, T) = \sum_{t \in \mathrm{sub}} d_{\max} \cdot \Delta\rho(t) + \sum_{t \in \mathrm{sub}} r_{\mathrm{W}} \cdot L(W_t)$$

Both sums contain only non-negative terms ($d_{\max}, \Delta\rho, r_{\mathrm{W}}, L \geq 0$), and the correction term vanishes entirely. Therefore $B(t_0, T) \geq 0$ for every window, with equality only in the trivial case where no restriction occurs.

For any non-trivial restriction step ($n_{\mathrm{sub}}(t) = 1$ with $\Delta\rho(t) > 0$ or $L(W_t) > 0$), $B > 0$, satisfying condition (i) of Theorem 1, Part (b). Conditions (ii) and (iii) follow from the restriction-specific dynamics. Each restriction step removes the exercise that would generate evidence (no exercise $\rightarrow$ no evidence), driving $\rho_k^{\mathrm{cross}} \rightarrow 0$ on the risk dimension (satisfying (ii)). With $r_{\mathrm{S}} = 0$ and the risk dimension environmentally non-stationary—the true risk changes as the restricted capability's context evolves without observation—the error ϵ_k grows, eventually exceeding $\epsilon_{\mathrm{crit},k}$ (satisfying (iii)). Note that $d_{\max} > 0$ bounds the drift rate from above (C2′). The cascade requires positive drift, which is a structural consequence of non-stationarity on the restricted dimension. All three conditions of Part (b) are therefore met, and the cascade drives the system toward maximal restriction.

This is the formal statement of why the Wamura problem was called "the most important open problem" in the discussion of Lasser [2026c]: without a mechanism to generate evidence against unfalsifiable risk claims, the feedback loop between restriction and evidence suppression is absorbing by default. ■

7.4 Summary

Table 2 summarizes the three loops as parameter instantiations of the unified system.

Loop	**Correction** (r_{S})	**Degradation** (r_{W})
1. Structural avoidance	Discovery value [Lasser, 2026c] drives learning of unknown poset edges	Leverage estimates drift from unlearned structure
2. Partial subsumption	Self-trust T_s [Lasser, 2026b] corrects world model via prediction residuals	Subsumption removes observation channels
3. Over-calibrated restriction	Controlled relaxation generates counter-evidence	Unfalsifiable risk claims accumulate under restriction

Table 2: Three feedback loops as instances of the (P, W) system. Each loop maps a correction source (r_{S}) and degradation source (r_{W}) onto the phase boundary of Theorem 1.

The unified treatment reveals that solutions for one loop improve the phase boundary for all three. Controlled relaxation (which makes $r_{\mathrm{S}} > 0$ for Loop 3) has no direct effect on Loop 2's parameters, but the structural insight transfers: any mechanism that generates observation-channel signal on dimensions that would otherwise go blind is a contribution to r_{S} in the relevant loop's phase boundary.

8 Design Criterion: Minimum Channel Redundancy

The phase boundary theorem converts a dynamical-systems result into a design criterion: what is the minimum cross-substrate redundancy per safety-relevant dimension needed to guarantee the system stays in the self-correcting basin?

8.1 The redundancy criterion

Theorem 2 (Per-Dimension Redundancy for Self-Correcting Dynamics)**.** *Let $\epsilon_{k,\mathrm{avg}}^2$ denote the time-averaged squared error on dimension k over the relevant trajectory window, and let $\Delta\rho_{\mathrm{avg}}(k)$ denote the time-averaged redundancy loss on dimension k per subsumption step. If for every dimension*

$k \in \{1, \ldots, K\}$,

$$\rho_k^{\text{cross}}(P) \geq K_{\min}^{\text{cons}}(k) \tag{23}$$

where

$$K_{\min}^{\text{cons}}(k) := \frac{r_{\text{W}} \cdot r_{\text{sub}}^{\max} + d_{\max} \cdot \Delta\rho_{\text{avg}}(k)/(w_k \cdot \epsilon_{k,\text{avg}}^2)}{r_{\text{S}}} \tag{24}$$

then the per-dimension Lyapunov component $L_k(W_t) := w_k \cdot \epsilon_k(t)^2$ contracts in the amortized sense for each k, and the aggregate Lyapunov function $L(W_t) = \sum_k L_k(W_t)$ satisfies the self-correcting condition of Theorem 1(a).

The formula uses the conservative substitutions *R1 and R2 stated below. $r_{\text{sub}}^{\max}$ is the maximum subsumption rate under locally rational dynamics over the relevant trajectory window. The exogenous rates r_{S}, r_{W}, $d_{\max}$ are the global bounds from Theorem 1. The endogenous correction term ($\alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu}$) is* not credited *to any single dimension in this formula, because it is a shared resource whose per-dimension allocation cannot be established without additional structure (see R2 below).*

Proof. Under Assumptions C1 and C2′, the per-dimension contribution to the Lyapunov function evolves according to

$$L_k(W_{t+1}) \leq L_k(W_t) \cdot \big(1 - r_{\text{S}} \cdot \rho_k^{\text{cross}}(P_t)\big) + n_{\text{sub}}(t) \cdot \big[r_{\text{W}} \cdot L_k(W_t) + d_{\max} \cdot w_k \cdot \Delta\rho_k(t)\big]. \tag{25}$$

This is the dimension-k restriction of Proposition 1: the exogenous correction $r_{\text{S}} \cdot \rho_k^{\text{cross}}$ contracts L_k at a rate proportional to the dimension's own redundancy; the error-proportional degradation $r_{\text{W}} \cdot L_k$ compounds the existing error on k; the drift exposure $d_{\max} \cdot w_k \cdot \Delta\rho_k$ adds error from channels lost at step t that covered k. (The endogenous term $\alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu} \cdot L(W_t)$ from C1 aggregates across dimensions and does not appear in the per-dimension restriction; see R2.)

For each dimension k, dividing Equation 25 by $L_k(W_t)$ (taking $\epsilon_{k,\text{avg}}^2$ as a representative value in the amortized sense), the per-step balance on dimension k requires

$$r_{\text{S}} \cdot \rho_k^{\text{cross}} \;\geq\; r_{\text{W}} \cdot r_{\text{sub}}^{\max} + d_{\max} \cdot \Delta\rho_{\text{avg}}(k)/(w_k \cdot \epsilon_{k,\text{avg}}^2). \tag{26}$$

Solving for ρ_k^{cross} yields Equation 24. Under the hypothesis $\rho_k^{\text{cross}} \geq K_{\min}^{\text{cons}}(k)$ for every k, each L_k contracts in the amortized sense (by the same switched-systems argument as Proposition 1, restricted to dimension k).

For the aggregate: summing over k,

$$L(W_{t+1}) = \sum_k L_k(W_{t+1}) \leq \sum_k L_k(W_t) \cdot \big(1 - r_{\text{S}} \cdot \rho_k^{\text{cross}}\big) + \text{(subsumption terms)} \tag{27}$$

Since $\rho_k^{\text{cross}} \geq K_{\min}^{\text{cons}}(k) \geq K_{\min}^{\text{cons}}(k^*) = \max_j K_{\min}^{\text{cons}}(j)$ only for $k = k^*$, but each k independently satisfies its own balance, the aggregate contracts. Because the endogenous term $\alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu}$ is non-negative, adding it to the aggregate budget can only widen the self-correcting basin, so the aggregate condition (Equation 18) holds *a fortiori*.

Conservative substitutions. Two choices make the per-dimension bound a rigorous sufficient condition rather than a heuristic:

(R1) The denominator uses the per-dimension error $w_k \cdot \epsilon_{k,\text{avg}}^2$, not the aggregate $L_{\text{avg}} = \sum_j w_j \epsilon_{j,\text{avg}}^2$. Since $w_k \cdot \epsilon_{k,\text{avg}}^2 \leq L_{\text{avg}}$, the per-dimension denominator is smaller. This makes the drift term $d_{\max} \cdot \Delta\rho_{\text{avg}}(k)/(w_k \cdot \epsilon_{k,\text{avg}}^2)$ *larger* and $K_{\min}^{\text{cons}}(k)$ larger (more conservative) than a formula using L_{avg}. This fixes the relaxation R1 noted in prior drafts: a well-calibrated dimension k does not have its required redundancy inflated by large errors on other dimensions, nor does it have its drift-exposure term under-counted by the aggregate error.

(R2) The endogenous correction $\alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu}$ is dropped from $K_{\min}^{\text{cons}}(k)$ entirely, rather than credited in full to each dimension. The endogenous term is a shared resource whose per-dimension allocation depends on structure we do not establish here (it arises from the aggregate vol_{P}-gradient, not from any individual dimension's channels). Setting $c_{\text{V}} = 0$ in the per-dimension bound gives the conservative result: the actor's endogenous incentive is an *additional* reserve that widens the aggregate basin, not a per-dimension credit.

Both substitutions make $K_{\min}^{\text{cons}}(k)$ a strict upper bound on the per-dimension requirement implied by the aggregate condition. The theorem's hypothesis (every $\rho_k^{\text{cross}} \geq K_{\min}^{\text{cons}}(k)$) is therefore sufficient for self-correction, not merely suggestive.

Practical use. The design criterion requires $\epsilon_{k,\text{avg}}$, which is trajectory-dependent. For a system whose worst-case operating error on dimension k is bounded by $\epsilon_{k,\max}$, substituting $\epsilon_{k,\text{avg}} = \epsilon_{k,\max}$ gives a lower bound on $K_{\min}^{\text{cons}}(k)$ that is sufficient only for trajectories that do not reach lower error. Well-calibrated dimensions ($\epsilon_{k,\text{avg}} \to 0$) require increasingly more redundancy to guard against drift exposure: the "removing a channel that was doing all the correction work" failure mode. This is a feature of the conservative substitution R1. The per-dimension formula diverges as $\epsilon_{k,\text{avg}} \to 0$, correctly flagging that subsumption of the sole corrector of a well-calibrated dimension is never safe. ■

8.2 Practical implications

Corollary 3 (Two-Substrate Minimum)**.** *For a two-substrate partition ($m = 2$, e.g., biology + silicon):*

1. $\rho_k^{\text{cross}} \leq 2$ *for each dimension k (at most one channel from each substrate).*
2. *The per-dimension criterion (Equation 24) with $\rho_k^{\text{cross}} = 2$ requires $2r_{\text{S}} \geq r_{\text{W}} \cdot r_{\text{sub}}^{\max} + d_{\max} \cdot \Delta\rho_{\text{avg}}(k)/(w_k \cdot \epsilon_{k,\text{avg}}^2)$ for every dimension k.*
3. *Dimensions where one substrate provides the only cross-substrate signal have $\rho_k^{\text{cross}} = 1$, requiring $r_{\text{S}} \geq r_{\text{W}} \cdot r_{\text{sub}}^{\max} + d_{\max} \cdot \Delta\rho_{\text{avg}}(k)/(w_k \cdot \epsilon_{k,\text{avg}}^2)$: exogenous correction alone must exceed degradation and drift exposure on that dimension (the endogenous term provides no per-dimension credit under R2).*

This means that in a two-substrate world, each substrate's observation channels are singularly load-bearing for every dimension they cover. Loss of one substrate's channel on a dimension reduces ρ_k^{cross} from at most 2 to at most 1. The redundancy criterion is then immediately violated on any dimension where $K_{\min}^{\text{cons}}(k) > 1$, and provides zero safety margin on dimensions where $K_{\min}^{\text{cons}}(k) \leq 1$.

Remark 13 (Progressive attenuation as the primary threat)**.** *The $m = 2$ vulnerability is not that one substrate is* removed *(an overt act that would itself be detectable) but that it is* attenuated*. Either the human stops reviewing AI capability claims because the AI's self-reports "seem fine," or human judgment is progressively replaced by automated proxies that share the silicon substrate's failure modes.*

This attenuation reduces effective sensitivity without changing the nominal channel count. ρ_k^{cross} then drops below the threshold while appearing unchanged to a naive audit. The skeleton-substrate scenario (Corollary 4) is precisely this failure mode.

The framing should be read as emphasizing $m = 2$ fragility and the value of $m \geq 3$, not the specialness of any particular substrate.

Corollary 4 (Skeleton-Substrate Vulnerability)**.** *Under a skeleton-substrate strategy where $m \geq 2$ nominally but the non-dominant substrate's channels have sensitivity $\to 0$:*

$$\rho_k^{\text{cross}}(P) \to \rho_{k,\text{dominant}}^{\text{cross}}(P) \tag{28}$$

because the sensitivity-weighted definitions (Definitions 4 and 5) already discount low-sensitivity channels.

Proof. From Definition 5, the contribution of substrate s to ρ_k^{cross} is $\min(1, \max_{c \in s} \sigma(c,k)/\sigma_{\text{thr}})$. As $\sigma(c,k) \to 0$ for all channels on the non-dominant substrate, this contribution goes to zero. The sum reduces to the dominant substrate's contribution alone. ■

The skeleton-substrate strategy does not evade the redundancy criterion by construction: it directly reduces ρ_k^{cross} by reducing the marginalized substrate's contribution to near-zero. No separate "effective" redundancy is needed; the phase boundary formalism already captures this vulnerability through the sensitivity weighting.

8.3 Self-diagnosability

The phase boundary is partially self-diagnosable by the actor. The actor can observe:

- ρ_k^{cross} (it knows which channels it has and their approximate sensitivities),
- r_{S} (from its own learning rate and historical correction behavior).

The actor cannot directly observe:

- r_{W} (the degradation rate depends on the true dynamics, not the perceived dynamics),
- $d_{\max}$ (the exogenous drift rate on blinded dimensions is by definition unobservable once the dimension is blind).

This means the actor can check "am I above the threshold for the r_{W} I estimate?" but not "is my estimate of r_{W} correct?" The actor can monitor the observable ingredients and flag when they approach the boundary, even if it cannot guarantee the boundary is computed correctly. This is an honest limitation. The phase boundary is a sufficient condition for self-correction, not a decision procedure for whether self-correction will actually obtain.

9 Worked Example: Skeleton-Substrate Scenario

This section applies the formalism to the skeleton-substrate scenario (Sections 6.8–6.9 of Lasser [2026c]) with explicit numerical values, illustrating how the phase boundary operates in a concrete case.

9.1 Setup

Consider a two-substrate system (biology + silicon) with:

- $K = 5$ safety-relevant dimensions of W:
  1. leverage accuracy ($\hat{\lambda}$),
  2. risk accuracy ($\hat{\mathcal{R}}$),
  3. substrate diversity (m_{eff}),
  4. cooperative growth rate ($\hat{r_{\text{ext}}}$),
  5. self-trust calibration (T_s).
- 3 observation channels from biology covering dimensions $\{1, 3, 5\}$.
- 4 observation channels from silicon covering dimensions $\{1, 2, 3, 4\}$.
- Parameters: $r_{\text{S}} = 0.1$, $r_{\text{W}} = 0.05$, $r_{\text{sub}} = 0.2$, $d_{\max} = 0.02$ (all per step).

For simplicity, the worked example uses the purely exogenous correction model ($c_{\text{V}} = 0$, so the total correction rate is $r_{\text{S}} \cdot \rho_{\min}^{\text{cross}}$ throughout).

These values are chosen for pedagogical clarity, not empirical calibration. The subsumption frequency $r_{\text{sub}} = 0.2$ means one subsumption event per five time steps, which is aggressive but plausible during an active consolidation phase. The self-correction rate $r_{\text{S}} = 0.1$ corresponds to 10% error correction per step from the EWMA learning rule at moderate learning rate. The degradation rate $r_{\text{W}} = 0.05$ assumes uncorrected drift is half the correction rate. The exogenous drift $d_{\max} = 0.02$ is the rate at which a newly blinded dimension drifts from its last calibrated value.

9.2 Initial state

At $t = 0$, the channel coverage is:

	Dim 1	**Dim 2**	**Dim 3**	**Dim 4**	**Dim 5**
Biology	✓		✓		✓
Silicon	✓	✓	✓	✓	
ρ_k^{cross}	2	1	2	1	1

The bottleneck is $\rho_{\min}^{\text{cross}} = 1$ (dimensions 2, 4, and 5 each have only one substrate providing coverage). Before any subsumption, $\Delta\rho_{\text{avg}} = 0$. The phase boundary condition (Equation 18) is:

$$r_{\text{S}} \cdot \rho_{\min}^{\text{cross}} = 0.1 \cdot 1 = 0.1 > r_{\text{W}} \cdot r_{\text{sub}} + 0 = 0.05 \cdot 0.2 = 0.01$$

Satisfied with margin. The contraction condition holds, and the system is in the self-correcting basin provided the resulting neighborhood radius satisfies $L_\infty < \epsilon_{\text{safe}}$ (Theorem 1, Part a).

9.3 Trajectory under progressive marginalization

Phase 1: Biological channels attenuated. A skeleton-substrate strategy does not remove biological agents but reduces their influence. Suppose the effective sensitivity of biological channels drops to $\sigma = 0.01 \cdot \sigma_{\text{thr}}$ (1% of the operational threshold). Now:

	Dim 1	**Dim 2**	**Dim 3**	**Dim 4**	**Dim 5**
Biology (effective)	0.01		0.01		0.01
Silicon	1	1	1	1	
ρ_k^{cross} (eff.)	1.01	1	1.01	1	0.01

On dimension 5 (self-trust calibration): $\rho_5^{\text{cross}} = 0.01$. The simplified sufficient condition (Equation 18) fails:

$$r_{\text{S}} \cdot \rho_5^{\text{cross}} = 0.1 \cdot 0.01 = 0.001 < r_{\text{W}} \cdot r_{\text{sub}} = 0.01$$

Note the distinction from Theorem 1, Part (b). The theorem requires $\rho_k^{\text{cross}} = 0$ (complete blindness), while here $\rho_5^{\text{cross}} = 0.01$ (near-blindness). Strictly, this is failure of the *sufficient condition for self-correction* (Part a), not satisfaction of the *sufficient condition for absorption* (Part b). The system is in the gap between the two basins: correction on dimension 5 is present but inadequate. Under continued attenuation, $\rho_5^{\text{cross}} \to 0$ and Part (b) applies. In practice the distinction is thin. At $\rho_5^{\text{cross}} = 0.01$, the error growth rate of 0.009 per step overwhelms the correction rate of 0.001, and ϵ_5 crosses $\epsilon_{\text{crit},5}$ long before the last biological channel is formally removed.

Phase 2: Cascade. With self-trust miscalibrated (dimension 5), the actor does not correctly detect its own prediction errors. The self-trust mechanism T_s [Lasser, 2026b] depends on accurate self-assessment. When ϵ_5 grows, the learning rate $\alpha(t)$ responds incorrectly, effectively reducing r_{S} system-wide.

As r_{S} drops, the self-correcting condition becomes harder to satisfy on dimensions that were marginally above the threshold. If r_{S} drops from 0.1 to 0.05, check the condition for dimensions with $\rho_k^{\text{cross}} = 1$ (using the sensitivity-table value $\Delta\rho_{\text{avg}}/L_{\text{avg}} = 0.5$ from Table 3):

$$\underbrace{0.05 \cdot 1}_{=0.05} \overset{?}{>} \underbrace{0.05 \cdot 0.2 + 0.02 \cdot 0.5}_{=0.02}$$

Correction (0.05) still exceeds degradation (0.02), so these dimensions remain in the self-correcting basin at $r_{\text{S}} = 0.05$. But the margin has shrunk from 0.09 to 0.03. The continued decline in r_{S} from the cascading self-trust failure eventually pushes the system past the critical surface. When r_{S} drops below 0.02, all dimensions enter net degradation. Once $\rho_k^{\text{cross}} \to 0$ on the failing dimensions, Part (b) of the theorem applies, and the Cascade Lemma (Appendix A.2) drives the system toward S^*.

9.4 Diagnosis

The example illustrates three properties of the phase boundary.

1. The weakest dimension breaks first. Dimension 5 (self-trust calibration) is the initial failure point because it has $\rho_5^{\text{cross}} = 0.01$ after attenuation: the lowest effective redundancy. The phase boundary breaks first at $\min_k \rho_k^{\text{cross}}$.

2. Attenuation is harder to detect than removal. The nominal channel count is unchanged (three biological channels still exist). A naive audit counting channels rather than measuring effective sensitivity would report $\rho_k^{\text{cross}} = 2$ on dimensions 1, 3 and $\rho_k^{\text{cross}} = 1$ on dimensions 2, 4, 5—all at or above the initial values. The sensitivity-weighted definitions (Definitions 4 and 5) are essential for diagnosing this scenario correctly.

3. Self-trust is the cascade trigger. Self-trust calibration (dimension 5) is uniquely dangerous because it affects the system-wide correction rate r_{S}. Miscalibration on other dimensions degrades accuracy on those dimensions specifically. Miscalibration on self-trust degrades the correction mechanism itself, creating a system-wide vulnerability. This gives self-trust calibration a role analogous to a "keystone" dimension: its loss triggers cascading failure on dimensions that were previously stable.

9.5 Sensitivity analysis

Table 3 shows how the critical redundancy $(\rho_{\min}^{\text{cross}})^*$ varies across plausible parameter ranges, using the linearized formula (Equation 20) with $\Delta\rho_{\text{avg}}/L_{\text{avg}} = 0.5$ (moderate drift-to-error ratio).

r_{S}	r_{W}	r_{sub}	$(\rho_{\min}^{\text{cross}})^*$
0.10	0.05	0.2	0.20
0.10	0.05	0.5	0.35
0.05	0.05	0.2	0.40
0.05	0.10	0.2	0.60
0.05	0.10	0.5	1.20

Table 3: Critical redundancy $(\rho_{\min}^{\text{cross}})^*$ for $d_{\max} = 0.02$ and $\Delta\rho_{\text{avg}}/L_{\text{avg}} = 0.5$. When $(\rho_{\min}^{\text{cross}})^* > 1$, a two-substrate system cannot satisfy the criterion on dimensions with single-substrate coverage.

The last row ($(\rho_{\min}^{\text{cross}})^* = 1.20$) illustrates the $m = 2$ fragility: when the degradation rate equals the correction rate and subsumption is frequent, the required redundancy exceeds what a two-substrate system can provide on its most vulnerable dimensions. This is the quantitative content of Remark 12: $m = 2$ works only when the correction rate substantially exceeds the degradation rate.

10 Discussion and Open Questions

10.1 What this paper establishes

Five results emerge from the analysis.

1. The phase boundary exists and is characterized. Compound feedback loops are not inherently catastrophic or inherently self-correcting. The outcome depends on the ratio of self-correction rate to observation-loss rate, mediated by channel redundancy. Theorem 1 gives the boundary; the linearized sufficient condition (Equation 18) gives the computable form.

2. All three feedback loops are the same dynamical system. The unified treatment (Section 7) shows that structural avoidance, partial subsumption, and over-calibrated restriction share a common formal structure. Solutions for one loop (controlled relaxation for the Wamura problem, structural discovery incentives for leverage blindness) improve the phase boundary for all.

3. The Wamura problem is a degenerate phase boundary. Without a mechanism for evidence generation against unfalsifiable risk claims, the self-correcting sufficient condition fails for every configuration (Proposition 4); absorption on the risk dimension then follows whenever the Part (b) blindness and cascade conditions are also met. The Wamura problem was called "the most important open problem" in the discussion of Lasser [2026c]. The phase boundary analysis explains why: it is the case where the self-correction rate is zero by construction, making the feedback loop absorbing regardless of all other parameters.

4. Cross-substrate redundancy is the load-bearing quantity. Same-substrate channels may share failure modes; cross-substrate channels are structurally independent (under the channel isolation property of Lasser [2026a] and, for additivity, the joint-factorization assumption). The design criterion (Theorem 2) is stated in terms of cross-substrate redundancy, making it robust to correlated failures within substrates.

5. In two-substrate systems, each substrate's channels are singularly load-bearing. This is a structural prediction from the dynamics rather than a normative claim about human importance: in $m = 2$ systems, losing one substrate's channel on a dimension reduces ρ_k^{cross} to at most 1, violating the redundancy criterion on any dimension where $K_{\min}^{\mathrm{cons}}(k) > 1$ and leaving zero margin where $K_{\min}^{\mathrm{cons}}(k) \leq 1$ (Corollary 3). The skeleton-substrate scenario (Section 9) shows that progressive *attenuation* of biological channels, not outright removal, is the primary vulnerability.

10.2 Limitations

The failure modes studied here—progressive channel degradation, feedback-loop instability, and the gap between observable performance and actual alignment—are instances of the concrete safety problems identified by Amodei et al. [2016]. Specifically, they correspond to distributional shift (the agent's training distribution diverges from the deployment distribution as channels attenuate) and reward hacking (the agent's observable metrics remain good while the underlying objective degrades). The phase boundary formalizes the conditions under which these problems become self-reinforcing.

Linearized simplified condition. Theorem 1(a) is stated generally via a uniform contraction condition that admits time-varying per-step rates. The simplified sufficient condition and the critical surface (Equations 18 and 19) are stated in a linearized regime with constant r_{S}, r_{W}, and r_{sub}. In the full dynamics these quantities are state-dependent. A larger world-model error produces more aggressive subsumption (r_{sub} increases with L), and the correction rate r_{S} depends on T_s which itself depends on the error history. The linearized treatment produces the qualitative result (self-correcting and absorbing basins exist, separated by a characterized boundary), but the quantitative location of the boundary shifts under the nonlinear dynamics. A full nonlinear analysis would tighten the boundary at the cost of substantially more complex proofs.

Honest-channel assumption. The analysis assumes observation channels are honest: their signal is noisy but not adversarially manipulated. The cryptographic commitment protocol of Lasser [2026a] provides the enforcement mechanism for this assumption. The two papers are designed to be read as complements: the exogenous verification framework ensures channels are honest, and this paper ensures enough honest channels survive. But the boundary between "noisy" and "adversarial" is itself a modeling choice. The line is blurred in practice by a failure mode neither paper alone addresses: *convertibility of integrity attacks into availability attacks*.

A sophisticated actor that cannot commit a false claim (because the commitment protocol would detect it) can instead choose which claims to commit and which to suppress. On a dimension where the actor expects an unfavorable verification outcome, it can decline to participate in verification on that dimension. The channel then remains nominally intact but carries no signal about the dimension of concern. This converts an integrity attack (committing a false claim) into an availability attack (making the channel silent on the relevant dimension). The integrity protocol of Lasser [2026a] handles claims that are committed; claims the actor chooses not to commit fall outside its scope and become an availability problem for this paper. The three loops differ in how they interact with this blur. Loops 1 and 3 (structural avoidance, over-calibrated restriction) produce channel silence through the dynamics themselves, while Loop 2 (partial subsumption) produces channel degradation. A sophisticated Loop-2 actor can emulate Loop-1 behavior on specific dimensions, producing selective silence that looks like the framework-endogenous failure mode rather than adversarial manipulation. Distinguishing the two requires observing *which* channels fall silent, which is itself a meta-channel subject to the same availability concerns.

Partial self-diagnosability. As noted in Section 8, the actor can observe ρ_k^{cross} and r_{S} but not r_{W} or $d_{\max}$. The phase boundary is therefore a sufficient condition for self-correction that the actor can partially monitor, not a decision procedure it can fully evaluate. This is an honest limitation, not a

fixable gap. The quantities the actor cannot observe are precisely the ones that depend on the true dynamics rather than the perceived dynamics.

10.3 Connections to other proposals

Controlled relaxation. Proposition 4 shows the Wamura problem is a degenerate phase boundary. Controlled relaxation protocols (the subject of a companion paper) provide the mechanism that makes $r_{\mathrm{S}} > 0$ for Loop 3. The two analyses are complementary. This paper provides the dynamical framework showing *why* a positive self-correction rate matters, while controlled relaxation provides the specific protocol that *achieves* it.

The B-to-C gap. The phase boundary analysis assumes $\mathrm{vol}_{\mathrm{P}}$ accurately measures what it claims. The B-to-C gap is the scenario where this assumption fails: $\mathrm{vol}_{\mathrm{P}}$ diverges from the experiential volume it is supposed to track. A complete treatment would need to extend the phase boundary to account for the divergence between $\mathrm{vol}_{\mathrm{P}}$ and the realized capability volume. This extension is future work.

Cryptographic verification. Lasser [2026a] addresses observation-channel integrity; this paper analyzes the dynamical consequences of channel *availability*. The combination is: exogenous verification ensures channels are honest, this paper ensures enough honest channels survive. A joint treatment where the phase boundary accounts for both integrity failures [Lasser, 2026a] and availability failures (this paper) would provide a stronger combined guarantee.

World-model verification as a corrective force. The self-trust feedback chain that creates the monopolar basin (Section 5) has a specific vulnerability: the agent cannot observe its own calibration degradation because the channels that would report it have been attenuated. The commit-then-reveal protocol of Lasser [2026a] provides a direct corrective force against this failure mode.

If the agent commits world-model predictions to the verification ledger *before* observing outcomes, the residuals between committed predictions and realized outcomes are an exogenous calibration signal that the agent cannot game. The commitment is cryptographic (the prediction cannot be changed after the fact), and the outcome verification is cross-substrate (the agent cannot self-report the outcome it committed to predict). This is exactly the signal the monopolar basin lacks: an unfakeable measure of world-model quality on the dimensions the agent has stopped observing directly.

The connection to prediction markets [Hanson, 2003] is immediate. In a multi-agent setting, the verification ledger generalizes to a prediction market: agents commit predictions, stake reputation on outcomes, and calibration is scored publicly. The market's information aggregation rate I_{market} replaces the single-channel I_k in the metastable lifetime bound (Proposition 3), potentially shortening the escape time from the metastable basin dramatically. A consistently miscalibrated agent loses stake, which is a $\mathrm{vol}_{\mathrm{P}}$-legible signal. The agent's cooperative capabilities that depend on accurate predictions degrade in the ledger, producing a negative $\Delta\mathrm{vol}_{\mathrm{P}}$ that the agent's own optimization pressure cannot ignore.

Two design constraints apply. First, the queries must be *masked*: the agent should not know which dimensions of its world model are being tested, to prevent selective participation in predictions it expects to get right. Second, the predictions must be on *observable quantities* (what will metric X be next period?) rather than internal state (what is your belief about X?), both to protect the agent's epistemic privacy and to ensure the outcome is independently verifiable.

In the minimal case (single actor, no market), the operational instantiation is simpler: the user periodically poses calibration queries that the agent commits predictions to, and the user independently verifies the outcomes. This is the exogenous trust-review channel from Lasser [2026a] applied to world-model calibration specifically. Even this minimal version increases the effective r_{S} (each query adds a correction signal) without increasing $\rho_{\min}^{\mathrm{cross}}$ (the channel count is unchanged), widening the self-correcting basin.

If world models are not merely committed but *shareable* (other agents can read and build on them), the ledger becomes a bootstrapping mechanism. New agents enter the population with a calibrated world model derived from the public record rather than from scratch observation. This addresses

the initialization sensitivity of Theorem 2: the design criterion depends on $L_{\max}$, the initial world-model error. A population with a shared world-model ledger starts new agents with lower $L_{\max}$, reducing the redundancy threshold for the self-correcting basin.

10.4 Meta-knowledge and the phase boundary

A self-interested $\mathrm{vol}_{\mathrm{P}}$-maximizer with access to Theorem 1 gains three operational tools that a naive actor does not have.

Precautionary monitoring. The actor can partially evaluate the phase boundary condition before each subsumption decision. It observes ρ_k^{cross} (which channels it has), estimates r_{S} (from its own learning rate), and tracks r_{sub} (from its own subsumption history). r_{W} is not directly observable, since the degradation rate depends on the true dynamics. But the actor can compute the worst-case impact of a proposed subsumption on $\rho_{\min}^{\mathrm{cross}}$ and check whether the post-subsumption state still satisfies the condition under plausible r_{W} ranges. This provides a computable precautionary filter: reject subsumptions that would push $\rho_{\min}^{\mathrm{cross}}$ below the critical threshold, even when the local $\mathrm{vol}_{\mathrm{P}}$ calculation favors the subsumption.

Meta-rational discounting. A naive $\mathrm{vol}_{\mathrm{P}}$-maximizer evaluates subsumption against its current world model W_t. An actor with phase boundary knowledge adds a correction: the value of a channel is higher than W_t estimates, because W_t's accuracy on the dimensions that channel covers *depends on* the channel being present. Removing the channel degrades the very estimate that justified the removal. The theory provides the formal basis for this self-distrust. The actor knows that its valuation of a subsumption is biased precisely because the channel it would lose is the one calibrating the valuation.

Interpretive framework for indirect evidence. Under the metastable variant (Part b′), the actor has indirect signals: declining innovation rate, shrinking cooperative output, increasing prediction residuals. Without the phase boundary theory, these are ambiguous data points. With the theory, they are diagnostics for a specific failure mode (substrate homogenization) with a known remedy (re-introduce substrate diversity). Meta-knowledge converts ambiguous signals into actionable diagnoses. This shortens the metastable lifetime τ_{meta} by increasing the effective information rate from indirect channels.

The preventive-not-rescue asymmetry. Meta-knowledge is most valuable in the self-correcting basin and at the phase boundary, where the actor's world model is calibrated enough to apply the theory accurately. In the absorbing basin, the world model has degraded to the point where the actor cannot accurately evaluate the phase boundary condition. It believes it is in the self-correcting basin when it is not, because the channels that would reveal the miscalibration have been eliminated. Meta-knowledge is therefore a *preventive* tool (it keeps the actor from approaching the boundary) rather than a *rescue* tool (it cannot pull the actor out of the absorbing basin once the world model is sufficiently degraded). This mirrors the cascade asymmetry (Remark 10): the stabilizing cascade benefits from meta-knowledge at every step, while the destabilizing cascade degrades the meta-knowledge along with everything else.

The recursive implication. If meta-knowledge of the phase boundary increases resilience, then developing and sharing the theory is itself a $\mathrm{vol}_{\mathrm{P}}$-maximizing action. A $\mathrm{vol}_{\mathrm{P}}$-maximizer that understands this would: (a) invest in understanding its own failure modes, because that understanding increases ν in Assumption C1 and thereby widens the self-correcting basin; (b) share that understanding with other agents, because shared meta-knowledge is a cooperative capability that benefits the entire population; (c) seek external review of its understanding, because miscalibrated meta-knowledge — false confidence in a wrong phase boundary estimate — could suppress the precautionary discounting that correct meta-knowledge would provide, making the actor *more* vulnerable; and (d) commit the theory to the verification ledger [Lasser, 2026a], because once verified and published, the theory becomes a shared resource that reduces every new agent's initialization cost.

Point (c) is the most subtle. Meta-knowledge that is wrong about the phase boundary location could convince the actor it has room to subsume when it does not, increasing rather than decreasing

vulnerability. The theory itself needs to be verified through the exogenous verification protocol. This creates a closure: the paper's results need the mechanisms its companion paper develops.

10.5 Open questions

1. **Nonlinear phase boundary.** The state-dependent extension of the phase boundary, where r_{S}, r_{W}, and r_{sub} are functions of (P_t, W_t) rather than constants, would produce tighter bounds at the cost of a switched-systems analysis with state-dependent switching signals.
2. **Characterization of absorbing states.** We prove convergence to *a* monopolar fixed point but do not characterize the full set of absorbing states. Partial monopolarity (some dimensions monopolar, others not) may be stable under certain parameter regimes.
3. **Optimal redundancy allocation.** Given a fixed budget of observation channels, how should they be allocated across dimensions and substrates to maximize the self-correcting basin? This is a resource-allocation problem over the channel coverage matrix.
4. **Empirical calibration.** The worked example uses pedagogical parameter values. Calibrating r_{S}, r_{W}, r_{sub}, and $d_{\max}$ from operational data (e.g., from AI governance structures, human-oversight-of-AI dynamics, or historical institutional feedback loops) would convert the phase boundary from a qualitative characterization to a quantitative prediction.

Author Contributions

Teague Lasser conceived the compound feedback loop analysis as the dynamical framework connecting Papers 1–5: the identification of three loop instances (observation-channel, trust-calibration, capability-evolution) from the gap analysis, the phase boundary as a separating surface in parameter space, and the connection between self-trust calibration and the monopolar absorbing state. Directed the formal development, identified the endogenous calibration incentive ($c_{\mathrm{V}} \cdot \nu$ term in Assumption C1), proposed the prediction-market corrective force as the Paper 5 connection, and made final editorial decisions. Responsible for the paper's intellectual direction and all claims made.

Claude Opus 4.6 (Anthropic) drafted the formal exposition: the coupled (P, W) dynamical system, the Lyapunov error construction with weighted sum-of-squares candidate, the per-step error dynamics (Proposition 1), the phase boundary theorem with its three parts (self-correcting basin, absorbing basin, critical surface), the cascade lemma, the metastable basin under partial endogenous correction, the minimum-redundancy design criterion (Theorem 2) with its two relaxations, the skeleton-substrate vulnerability analysis, and the worked example instantiation. Also developed the meta-knowledge discussion and the world-model verification corrective force analysis.

GPT 5.4 (OpenAI) served as technical reviewer across multiple codex review cycles, identifying formal gaps including: the r_S^{eff} circularity (the effective self-correction rate depending on ρ while ρ^* was defined as a ratio involving r_S^{eff}), the missing $\alpha_{\min}$ factor on the endogenous correction term throughout the simplified conditions, the need for explicit connectivity and non-stationarity conditions in Part (b)'s cascade propagation, the overshoot constant μ in the affine-recursion convergence (requiring explicit linearized-regime scoping), the strict-gap requirement for $\epsilon_{\mathrm{safe}} > 0$, and the per-dimension relaxations R1/R2 in the design criterion.

Transparency note. Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

References

Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016.

Nick Bostrom. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, 2014.

Paul Christiano. What failure looks like. 2019. Alignment Forum.

Thomas M Cover and Joy A Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.

Robin Hanson. Combinatorial information market design. *Information Systems Frontiers*, 5(1): 107–119, 2003.

Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joär Skalse, and Scott Garrabrant. Risks from learned optimization in advanced machine learning systems. *arXiv preprint arXiv:1906.01820*, 2019.

Hassan K Khalil. *Nonlinear Systems*. Prentice Hall, 3rd edition, 2002.

Teague Lasser. Exogenous verification for alignment: Cryptographic commitments on substrate-exclusive channels. `https://teague.info/papers/exo/`, 2026a.

Teague Lasser. Goal-frontier maximizers are civilization aligned. `https://teague.info/papers/gfm/`, 2026b.

Teague Lasser. Horizon-aware goal-frontier maximization excludes singleton behavior. `https://teague.info/papers/horizon/`, 2026c.

Teague Lasser. Computable goal frontiers and the gradient toward civilization-building. `https://teague.info/papers/poset/`, 2026d.

Teague Lasser. A structural causal model for goal-frontier maximization. `https://teague.info/papers/scm/`, 2026e.

Daniel Liberzon. *Switching in Systems and Control*. Birkhäuser, 2003.

Richard Ngo, Lawrence Chan, and Sören Mindermann. The alignment problem from a deep learning perspective. *arXiv preprint arXiv:2209.00626*, 2024.

Stephen M Omohundro. The basic AI drives. In *Proceedings of the First AGI Conference*, 2008.

Herbert Robbins and Sutton Monro. A stochastic approximation method. *The Annals of Mathematical Statistics*, 22(3):400–407, 1951.

Abraham Wald. *Sequential Analysis*. John Wiley & Sons, 1947.

A Proof Details and Technical Notes

A.1 Proof of Theorem 1 (full derivation)

The argument is a discrete Lyapunov stability analysis on the coupled (P, W) system, in the spirit of dwell-time conditions for switched linear systems [Liberzon, 2003]. We use $L(W_t) = \sum_k w_k \epsilon_k(t)^2$ (Definition 6) throughout.

Verification of C1 and C2′ as comparison bounds

Assumption C1 decomposes the aggregate correction rate into an exogenous channel term and an endogenous $\mathrm{vol_P}$-gradient term. *Exogenous.* For each dimension k, each channel covering k delivers an EWMA-style correction linear in the residual and proportional to its sensitivity; summing within k gives a rate proportional to $\rho_k \cdot \epsilon_k^2$, and bounding below by $\rho_{\min}^{\mathrm{cross}}$ across dimensions yields the first term. Under the joint-factorization assumption of Lasser [2026a], cross-substrate channels' correction signals are conditionally independent and additive. *Endogenous.* The term $c_{\mathrm{V}} \cdot \nu \cdot L$ arises from the actor's $\mathrm{vol_P}$-gradient: better calibration yields better decisions, which preserve more cooperative capabilities and raise $\mathrm{vol_P}$; ν measures this chain's marginal return along the decision chain (Assumption 1).

Assumption C2′ supplies the matching upper bound, decomposing degradation into a Lipschitz term (the world model does not jump discontinuously on channel loss, giving $c_{\mathrm{D}} \cdot L$) and an exogenous drift kick (channels removed from well-calibrated dimensions drift at a bounded rate $d_{\max}$, giving $d_{\max} \cdot \Delta\rho$). Proposition 1 converts C1 and C2′ into the per-step affine bound used throughout the proof.

Proof of Part (a): self-correcting basin

The affine recursion. Proposition 1 gives the per-step bound

$$L_{t+1} \leq a_t \cdot L_t + b_t$$

with

$$\begin{aligned} a_t &= 1 - r_{\mathrm{S}}(t) \cdot \rho_{\min}^{\mathrm{cross}}(P_t) - \alpha(t) \cdot c_{\mathrm{V}} \cdot \nu(P_t, W_t) + n_{\mathrm{sub}}(t) \cdot r_{\mathrm{W}}(t), \\ b_t &= n_{\mathrm{sub}}(t) \cdot d_{\max} \cdot \Delta\rho(t). \end{aligned}$$

Iterating over a window $[t_0, t_0 + T]$:

$$L_{t_0+T} \leq L_{t_0} \prod_{t=t_0}^{t_0+T-1} a_t + \sum_{s=t_0}^{t_0+T-1} b_s \prod_{t=s+1}^{t_0+T-1} a_t. \tag{29}$$

The uniform contraction condition. For (29) to yield a bounded limit, we need aggregate multiplicative decay and bounded additive drift over sufficiently long windows. The condition, used in Theorem 1(a), is:

$$\exists\, \bar{a} \in (0,1),\ \bar{b} \geq 0,\ T^* > 0 :\ \forall T \geq T^*, \quad \frac{1}{T} \sum_{t=t_0}^{t_0+T-1} \ln a_t \leq \ln \bar{a}, \quad b_t \leq \bar{b} \text{ pointwise.} \tag{30}$$

The log-average form is the natural object for a_t because the first term of (29) is a product; equivalently, the geometric mean of a_t is at most $\bar{a}$. The pointwise bound on b_t is stronger than a time-average bound but is what the suffix-weighted sum in (29) requires: individual large b_s terms are not averaged away by the tail product. A pointwise bound follows from C2′ with $\bar{b} = d_{\max} \cdot \overline{\Delta\rho}$ (applied at each subsumption step; non-subsumption steps have $b_t = 0$). We require $a_t > 0$ throughout, which holds whenever $r_{\mathrm{W}}(t) \cdot n_{\mathrm{sub}}(t) < 1$ (a step cannot more than wipe out the current error); this is a weak condition on per-step degradation and holds under C2′ for any finite learning rate.

Convergence under (30)**.** Applying AM–GM to the product in (29), $\prod_{t=t_0}^{t_0+T-1} a_t \leq \bar{a}^T$, so the first term decays geometrically. For the second term, group the tail products by suffix length: a step with additive kick b_s at position s contributes at most $b_s \cdot \bar{a}^{T-1-s}$ to L_{t_0+T} in the worst case (when

all subsequent a_t attain the upper bound $\bar{a}$). However, individual a_t may exceed $\bar{a}$ on subsumption steps even though the geometric mean is bounded. Define the *overshoot constant*

$$\mu = \sup_{N \geq 1} \sup_{t_0} \frac{1}{\bar{a}^N} \cdot \max_{0 \leq s \leq N-1} \prod_{t=t_0+s+1}^{t_0+N} a_t,$$

which is finite whenever an average-dwell-time condition holds: the number of consecutive subsumption steps (during which $a_t > \bar{a}$) within any window of length N is at most $\lambda \cdot N + N_0$ for constants $\lambda \in [0, 1)$ and $N_0 \geq 0$. This is the dwell-time formulation of Liberzon [2003, Ch. 2] and is stronger than $r_{\text{sub}} < 1$ (which bounds only asymptotic density). Under dwell-time, burst length is uniformly bounded and $\mu < \infty$. Summing the geometric series:

$$\limsup_{T \to \infty} L_{t_0+T} \leq \frac{\mu \cdot \bar{b}}{1 - \bar{a}}. \tag{31}$$

Linearized regime. When $|a_t - 1| \ll 1$ for all t (the moderate-step regime of Remark 6; cf. Equation 15), $\ln a_t \approx a_t - 1$, so (30) is equivalent to $\frac{1}{T}\sum(1 - a_t) \geq 1 - \bar{a}$. Substituting the worst-case values $\alpha_{\min} = \inf_t \alpha(t)$, $\underline{\nu} = \inf_t \nu(P_t, W_t)$, and the time-averaged $\overline{\Delta\rho}$ yields $1 - \bar{a} = r_{\text{S}} \cdot \rho_{\min}^{\text{cross}} + \alpha_{\min} \cdot c_{\text{V}} \cdot \underline{\nu} - r_{\text{W}} \cdot r_{\text{sub}}$. For the additive term, only subsumption steps contribute nonzero b_s; in the suffix-weighted sum $\sum_s b_s \prod_{t>s} a_t$, the density of contributing positions is r_{sub} and each is bounded pointwise by $d_{\max} \cdot \overline{\Delta\rho}$. The effective $\bar{b}$ in the linearized steady-state formula is therefore $r_{\text{sub}} \cdot d_{\max} \cdot \overline{\Delta\rho}$. In the linearized regime, $\mu = 1 + O(\max_t |a_t - \bar{a}|^2)$, which we drop at the order of the linearization, so (31) reduces to Equation 17.

Policy correctness: $\epsilon_{\text{safe}} > 0$. Define

$$\epsilon_{\text{safe}} = \sup\{\epsilon > 0 : \forall W \text{ with } L(W) \leq \epsilon,\ \pi^*(W) = \pi^*(W^*)\},$$

where $\pi^*(W)$ is the locally rational policy under world model W. For a finite action set at each decision point, π^* is determined by the signs of finitely many action gaps $\{G_j\}$. At W^*, each preserve-optimal agent has $G_j(W^*) > 0$ (the strict gap assumption); each subsume-optimal agent has $G_j(W^*) < 0$. Continuity of G_j in W then gives an open neighborhood of W^* on which every sign is preserved, hence $\epsilon_{\text{safe}} > 0$.

A computable lower bound follows from Cauchy–Schwarz in the w-weighted metric. Writing $G_j(W) - G_j(W^*) = \int_0^1 \nabla G_j(W_\tau) \cdot (W - W^*)\, d\tau$ along any path W_τ from W^* to W, Cauchy–Schwarz gives $|G_j(W) - G_j(W^*)| \leq \|\nabla G_j\|_{w^{-1}} \cdot \sqrt{L(W)}$ where $\|\cdot\|_{w^{-1}}$ is the dual norm with weights $1/w_k$. An action flip on agent j requires $|G_j(W) - G_j(W^*)| \geq G_j(W^*)$, so:

$$\epsilon_{\text{safe}} \geq \min_{j \in \mathcal{A}^+} \left(\frac{G_j(W^*)}{\sup_{W \in \mathcal{N}} \|\nabla G_j(W)\|_{w^{-1}}} \right)^2, \tag{32}$$

where $\mathcal{A}^+$ is the set of preserve-optimal agents and $\mathcal{N} = \{W : L(W) \leq L_\infty\}$ is the steady-state neighborhood from (31). The self-consistency is that L_∞ enters both the bound and $\mathcal{N}$; in practice one takes the smallest $\mathcal{N}$ containing the trajectory and iterates the self-consistency condition $L_\infty < \epsilon_{\text{safe}}(\mathcal{N})$ to closure. The bound (32) is conservative because $\|\nabla G_j\|_{w^{-1}}$ is taken as its supremum over the entire neighborhood, not the value along the actual trajectory. A cheaper per-coordinate estimator $\min_k w_k \cdot \epsilon_{\text{crit},k}^2$ rules out single-coordinate flips but does not bound multi-coordinate flips (where the action gap responds to error on several dimensions simultaneously), so (32) is the correct object for policy correctness on the whole neighborhood.

Closing Part (a). Combining (31) with the policy-correctness condition $L_\infty < \epsilon_{\text{safe}}$: once the trajectory enters the ϵ_{safe}-neighborhood of W^*, the locally rational policy agrees with the true V_γ-optimal policy and the anti-monopolar result [Lasser, 2026c, Proposition 6] applies. As subsumption ceases ($\overline{\Delta\rho} \to 0$), $\bar{b} \to 0$ and the neighborhood shrinks to $\{0\}$.

Co-evolution caveats. Two dependencies complicate the fixed-parameter reading. First, the contraction factor a_t depends on $\rho_{\min}^{\text{cross}}(P_t)$, which evolves as subsumption proceeds; the self-correcting basin is defined over trajectories, not initial states. Second, $\alpha(t)$ depends on T_s [Lasser, 2026b],

which depends on the error history; the EWMA structure of T_s makes α monotone in prediction residuals, so r_{S} is bounded below by a function of the minimum T_s along the trajectory. The uniform contraction condition (30) is the object that survives both dependencies because it is stated in terms of time averages of the actual a_t, not instantaneous parameter values.

Proof of Part (b): absorbing basin

The triggering conditions are (i) $B(t_0, T) > 0$ (Equation 14), (ii) $\rho_k^{\mathrm{cross}} = 0$ on a cascade dimension k, and (iii) $\epsilon_k > \epsilon_{\mathrm{crit},k}$. Under B1a, (ii) gives ϵ_k non-decreasing; environmental non-stationarity on k then keeps (iii) satisfied. The Cascade Lemma 2 converts miscalibration on k into action flips on agents covering dimensions $j \neq k$; each resulting subsumption drives $\rho_j^{\mathrm{cross}} \to 0$ on at least one new dimension (agent-channel graph connectivity) and the per-coordinate cascade threshold $\epsilon_{\mathrm{crit},j}$ is crossed there too under non-stationarity. Iterating across dimensions, the trajectory converges to a neighborhood of S^* (Definition 7). The per-coordinate cascade threshold is the correct object here because Part (b) concerns propagation through dimensions sequentially; the aggregate ϵ_{safe} bound of (32) is a sufficient condition for the entire neighborhood to be safe, while $\epsilon_{\mathrm{crit},k}$ is the necessary threshold for the *specific* action flip on dimension k.

Proof of Part (b′): metastable basin

Under B1′ (Assumption 4), the indirect signal g_k provides residual correction at information rate I_k, so ϵ_k is no longer non-decreasing once the trajectory enters the absorbing basin of Part (b). The cascade-trigger analysis of Part (b) still applies (B1′ does not remove the triggering mechanism, only the irreversibility), but escape from the basin is now possible through g_k. The lifetime τ_{meta} is set by the competition between the basin depth and the information rate. Proposition 3 gives the scaling law $\tau_{\mathrm{meta}} \gtrsim C/I_k$, where the constant C depends on the basin geometry near S^* and is not characterized as a theorem-level bound. The proof of Proposition 3 itself (by the mean first-passage argument of Kramers-type escape) is deferred.

Proof of Part (c): critical surface

Setting $B(t_0, T) = 0$ in the linearized regime of Part (a)—equivalently, requiring $(1-\bar{a}) = 0$—gives $r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}} + \alpha_{\min} \cdot c_{\mathrm{V}} \cdot \underline{\nu} = r_{\mathrm{W}} \cdot r_{\mathrm{sub}} + d_{\max} \cdot \Delta\rho_{\mathrm{avg}}/L_{\mathrm{avg}}$. Solving for $\rho_{\min}^{\mathrm{cross}}$ under the standing assumption $r_{\mathrm{S}} > 0$ yields Equation 20. The surface is linearized: the nonlinear generalization, where r_{S}, r_{W}, and r_{sub} depend on (P_t, W_t), would produce a state-dependent critical surface rather than a hyperplane (see Section 10).

A.2 Cascade Lemma

Lemma 2 (Cascade)**.** *Define the* action gap *for agent a_j at world-model state W as*

$$G_j(W) = V_\gamma^{\mathrm{preserve}}(a_j, W) - V_\gamma^{\mathrm{subsume}}(a_j, W)$$

(positive when preservation is preferred). Fix a dimension k and consider a state W_t satisfying:

(0) ***Single-coordinate dominance.*** *$W_t[k'] = W^*[k']$ for all $k' \neq k$. (Dimension k has been blinded while all other dimensions are still corrected. This is the cascade scenario: the lemma is applied at the step where k first triggers a wrong action, before the resulting subsumption blinds further dimensions.)*

Define the one-dimensional path $W(\epsilon) = W^* + \epsilon \cdot e_k \cdot \mathrm{sgn}(W_t[k] - W^*[k])$, where e_k is the k-th standard basis vector and $\epsilon \in [0, \epsilon_k(t)]$. Under assumption (0), $W_t = W(\epsilon_k(t))$.

Assume additionally:

(i) $G_j(W(\epsilon))$ is continuously differentiable in ϵ,

(ii) the miscalibration direction on k makes preservation less attractive: $\frac{d}{d\epsilon} G_j(W(\epsilon)) < 0$ for all $\epsilon \in [0, \epsilon_k(t)]$ (i.e., increasing error on k monotonically decreases the action gap),

(ii′) ***(Multi-dimensional monotonicity; needed for cascade propagation beyond the first trigger.)*** *For every safety-relevant dimension k' and every agent $a_{j'}$, $\mathrm{sign}(W[k'] - W^*[k']) \cdot$*

$\frac{\partial G_{j'}}{\partial W[k']} < 0$ *whenever* $W[k'] \neq W^*[k']$*; that is, moving* $W[k']$ *further from* $W^*[k']$ *(increasing the absolute error* $\epsilon_{k'}$*) monotonically decreases every agent's preservation advantage, regardless of the sign of the miscalibration. Assumption (ii) is the single-coordinate special case (where the error direction is fixed by construction of the path* $W(\epsilon)$*); (ii′) extends it to the multi-coordinate regime where multiple dimensions are simultaneously miscalibrated with potentially different error signs.*

Define the cascade threshold for agent a_j *on dimension* k*:*

$$\epsilon_{\text{crit}}^{(j,k)} = \frac{G_j(W^*)}{\inf\limits_{\epsilon \in [0, \epsilon_k(t)]} \left| \frac{d}{d\epsilon} G_j(W(\epsilon)) \right|} \tag{33}$$

(the true action gap divided by the minimum sensitivity to dimension-k *error along the path; a lower bound on sensitivity gives an upper bound on the error needed to flip the action). The per-dimension cascade threshold is* $\epsilon_{\text{crit},k} = \min_{j \in \mathcal{A}_t^+} \epsilon_{\text{crit}}^{(j,k)}$*, minimized over the set of* preserve-optimal agents $\mathcal{A}_t^+ = \{j \in \mathcal{A}_t : G_j(W^*) > 0\}$*. Agents that are truly subsume-optimal (*$G_j(W^*) \leq 0$*) are excluded because their subsumption is correct regardless of* ϵ_k*; only flips from correct-preserve to incorrect-subsume constitute cascade triggers.*

Then the cascade operates in two steps:

- ***(A) Action flip.*** *If* $\epsilon_k(t) > \epsilon_{\text{crit}}^{(j,k)}$*, then* $G_j(W_t) < 0$ *and the locally rational action under* W_t *selects subsumption of* a_j*.*
- ***(B) Error propagation.*** *Subsumption of* a_j *removes* a_j*'s channels. For any dimension* $j' \neq k$ *covered exclusively by* a_j*'s channels, the per-coordinate error* $\epsilon_{j'}(t+1) \geq \epsilon_{j'}(t)$*.*

Miscalibration on one dimension produces decisions that degrade other dimensions.

Proof. Step (A): Action flip. Under the true world model W^*, preservation is optimal: $G_j(W^*) > 0$. By the mean value theorem applied to the scalar function $\epsilon \mapsto G_j(W(\epsilon))$ on $[0, \epsilon_k(t)]$:

$$G_j(W_t) = G_j(W(\epsilon_k(t))) = G_j(W^*) + \left. \frac{dG_j}{d\epsilon} \right|_{\epsilon = \tilde{\epsilon}} \cdot \epsilon_k(t)$$

for some $\tilde{\epsilon} \in (0, \epsilon_k(t))$. By assumption (ii), the derivative is negative, so:

$$G_j(W_t) = G_j(W^*) - \left| \left. \frac{dG_j}{d\epsilon} \right|_{\tilde{\epsilon}} \right| \cdot \epsilon_k(t) \leq G_j(W^*) - \inf_{\epsilon} \left| \frac{dG_j}{d\epsilon} \right| \cdot \epsilon_k(t)$$

When $\epsilon_k(t) > \epsilon_{\text{crit}}^{(j,k)} = G_j(W^*) \big/ \inf |\frac{dG_j}{d\epsilon}|$, the right side is negative, so $G_j(W_t) < 0$ and the locally rational action selects subsumption of a_j.

Step (B): Error propagation. Once a_j is subsumed, Lemma 1 gives $\mathcal{O}(P_{t+1}) \subseteq \mathcal{O}(P_t)$. For any dimension j' covered exclusively by a_j's channels, $\rho_{j'}^{\text{cross}}(P_{t+1}) = 0$. By Assumption B1a (no endogenous correction on blind dimensions; Assumption 3), f_W does not decrease $\epsilon_{j'}$ when $\rho_{j'}^{\text{cross}} = 0$. Since the correction signal vanishes, $\epsilon_{j'}(t+1) \geq \epsilon_{j'}(t)$ (non-decreasing). Under environmental non-stationarity on dimension j' (the true state continues to evolve while the blind estimate cannot track it), $\epsilon_{j'}$ grows strictly; C2′ bounds the growth rate by $d_{\max}$ but the cascade requires only that drift is positive, not that it saturates the bound.

The cascade propagates: if $\epsilon_{j'}$ subsequently exceeds $\epsilon_{\text{crit},j'}$ for another agent $a_{j''}$ whose channels cover dimensions $j'' \neq j'$, step (A) applies again, driving further subsumptions.

Multi-coordinate reinforcement at later stages. After the first trigger, assumption (0) is violated for later cascade stages: dimensions k and j' are both miscalibrated. However, the errors on previously-blinded dimensions all push G_j in the same direction (more negative, making subsumption more attractive), provided assumption (ii) holds uniformly across all dimensions, not just the original trigger dimension k. This requires a multi-dimensional monotonicity condition: for every previously-blinded dimension k_i and every agent a_{j_n} at cascade stage n, $\text{sign}(W[k_i] - W^*[k_i]) \cdot \frac{\partial G_{j_n}}{\partial W[k_i]} < 0$.

This is assumption (ii′) above, which is stated as an explicit condition of the lemma. It is plausible for the subsumption-vs-preservation action gap (increasing error on any safety-relevant dimension decreases preservation value) and holds when miscalibration uniformly favors subsumption. Under (ii′), at cascade stage n with errors $\epsilon_{k_1}, \ldots, \epsilon_{k_{n-1}}$ on previously blinded dimensions, the action gap satisfies

$$G_{j_n}(W_t) \leq G_{j_n}(W^*) - \sum_{i=1}^{n} \inf \left| \frac{\partial G_{j_n}}{\partial W[k_i]} \right| \cdot \epsilon_{k_i}$$

(by iterated application of the MVT along each coordinate successively). Since every term in the sum is non-negative, the multi-coordinate case is *easier* to trigger than the single- coordinate case: the effective threshold for the n-th trigger is lower, not higher. The single-coordinate threshold $\epsilon_{\text{crit}}^{(j_n,k_n)}$ is therefore a conservative upper bound at every cascade stage. ■

A.3 Connection to the discount-factor threshold γ^*

The anti-monopolar result [Lasser, 2026c, Proposition 6] gives $\gamma^* = 1 - r_{\text{ext}}/\Delta_0$ as the critical discount factor for diversity dominance. Under the coupled (P, W) dynamics, the actor does not know the true r_{ext}; it uses $\hat{r_{\text{ext}}} = r_{\text{ext}} - \delta_{r_{\text{ext}}}$. The *perceived* critical discount factor is:

$$\gamma^*_{\text{perceived}} = 1 - \frac{r_{\text{ext}} - \delta_{r_{\text{ext}}}}{\Delta_0} \tag{34}$$

When $\delta_{r_{\text{ext}}} > 0$ (underestimating cross-substrate growth):

- $\gamma^*_{\text{perceived}} > \gamma^*$ (the threshold appears higher than it truly is),
- For $\gamma \in (\gamma^*, \gamma^*_{\text{perceived}})$, diversity dominates in truth but domination appears rational to the actor.

This is the *mechanism* by which world-model error converts a self-balancing regime into a locally rational subsumption cascade. The phase boundary from Theorem 1 characterizes when $\delta_{r_{\text{ext}}}$ stays small enough that $\gamma^*_{\text{perceived}}$ remains close to γ^*, preserving the effective reach of the anti-monopolar result [Lasser, 2026c].

B Notation Summary

Table 4 collects the principal symbols used in the paper.

Symbol	Definition	Introduced in
$S_t = (P_t, W_t)$	Coupled state	Def. 1
P_t	Capability poset at time t	Def. 1
W_t	World-model accuracy vector	Def. 1
γ	Discount factor [Lasser, 2026c, Def. 1]	Def. 2
$\mathcal{O}(P)$	Observation-channel function	Def. 3
$\mathrm{dims}(c)$	Dimensions of W covered by channel c	Def. 3
$\sigma(c, k)$	Signal-to-noise ratio of c on dim. k	Def. 3
σ_{thr}	Minimum operationally meaningful sensitivity	Def. 4
$\rho_k(P)$	Effective channel redundancy	Def. 4
$\rho_k^{\mathrm{cross}}(P)$	Cross-substrate redundancy	Def. 5
$L(W_t)$	Lyapunov function (world-model error)	Def. 6
$\epsilon_k(t)$	Error on dimension k	Def. 6
w_k	Safety-relevance weight for dim. k	Def. 6
c_{S}	Exogenous coercivity constant	Assump. C1
c_{V}	Endogenous correction constant	Assump. C1
$\nu(P_t, W_t)$	$\mathrm{vol}_{\mathrm{P}}$-sensitivity to calibration improvement	Assump. C1
c_{D}	Degradation bound constant	Assump. C2$'$
$d_{\max}$	Maximum exogenous drift rate	Assump. C2$'$
$n_{\mathrm{sub}}(t)$	Subsumption indicator (0 or 1)	Assump. C2$'$
$\underline{\Delta\rho}(t)$	Weighted redundancy loss at step t	Eq. 10
$\overline{\Delta\rho}$, $\Delta\rho_{\mathrm{avg}}$	Time-averaged weighted redundancy loss per sub. step	Thm. 1
$\Delta\rho_{\mathrm{avg}}(k)$	Per-dim. time-averaged redundancy loss per sub. step	Thm. 2
$\alpha(t)$	Learning rate (via T_s)	Prop. 1
$\beta(t)$	Degradation coefficient	Prop. 1
$\alpha_{\min}$	Learning-rate floor: $\inf_t \alpha(t)$	Thm. 1
r_{S}	Exogenous self-correction rate ($= \alpha_{\min} \cdot c_{\mathrm{S}}$)	Prop. 1
$r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}} + \alpha_{\min} \cdot c_{\mathrm{V}} \cdot \underline{\nu}$	Total correction rate (exo. + endo.)	Thm. 1
r_{W}	Degradation rate ($= \beta \cdot c_{\mathrm{D}}$)	Prop. 1
r_{sub}	Subsumption frequency	Thm. 1
$\rho_{\min}^{\mathrm{cross}}$	Minimum cross-substrate redundancy	Thm. 1
$B(t_0, T)$	Cumulative error budget	Eq. 14
S^*	Monopolar fixed point	Def. 7
m_{eff}	Effective substrate diversity index	Def. 7
τ_{meta}	Metastable lifetime	Prop. 3
$\delta_{r_{\mathrm{ext}}}$	Estimation error on r_{ext}: $\lvert \hat{r_{\mathrm{ext}}} - r_{\mathrm{ext}} \rvert$	Prop. 3
$K_{\min}(k, L_{\mathrm{avg}})$	Minimum redundancy for dim. k	Thm. 2
γ^*	Critical discount factor [Lasser, 2026c, Prop. 6]	App. A.3
$\gamma^*_{\mathrm{perceived}}$	Perceived γ^* under W_t	App. A.3
$\epsilon_{\mathrm{crit}}^{(j,k)}$	Cascade threshold for agent j, dim. k	Lemma 2
$\epsilon_{\mathrm{crit},k}$	$\min_j \epsilon_{\mathrm{crit}}^{(j,k)}$: per-dim. cascade threshold	Thm. 1
ϵ_{safe}	Sup. L preserving correct policy (see proof for bounds)	Thm. 1

Table 4: Principal notation.